Flight events from ADS-B

Version 4 — A-CDM milestones at twenty aerodromes

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Version 3 tested OPDI's published flight events against reference data for the first time and found that the two runway milestones it added, `ATOT` and `ALDT`, were the extreme samples of a detection window rather than measured instants. Version 4 replaces them. Nine A-CDM milestones — line-up, take-off roll, airborne, landing, touchdown, runway vacated, go-around and the two runway-crossing types — are now derived from one explicitly classified *runway traversal*, and the three that are instants are interpolated to the geometry that defines them rather than sampled near it. The vertical profile moves to EUROCONTROL PRU's published methodology: a 200 NM analysis radius, tops that relocate across a cruise-length level, and a rolling-window definition of level flight. Two silent unit defects are closed — the airport-layout gate and the level-segment floors were compared against uncorrected pressure altitude, so both moved with field elevation. Scoring is per aerodrome, over the twenty best-covered aerodromes in the network on three days of June 2026, with a column saying whether each aerodrome's reference times are second- or minute-resolution.

Each milestone has its own chapter: what it is, how it is detected, its parameters, what the reference data says, and what is known to be wrong with it. Retired milestones are not re-documented; each has one line in the breaking-changes chapter pointing at its successor.

This document may render before its measurement campaign has run. Every figure is produced by `benchmarks/regenerate_events_v4.py`; where an output is not yet staged the table says so in place rather than being omitted, and the provenance chapter lists what is present. A draft with visible holes is honest; a draft with numbers of unknown origin is not.

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1 The study in a page

OPDI turns OpenSky ADS-B state vectors into a published table of flight events: milestones with a 4D fix, a type, a source and an algorithm version. Version 3 scored them against EUROCONTROL APDF for the first time and reached two conclusions that shaped this version. The runway milestones it introduced, `ATOT` and `ALDT`, were **the extreme sample of a detection window** rather than a measured instant, and paid for it with a bias always in the same direction. And the runway they named was wrong about a third of the time on arrivals, because parallel runways were separated by a constant of the airport rather than by anything about the flight.

Version 4 is the answer to both, plus a third thing V3 could not see: several thresholds were being compared against **uncorrected pressure altitude**, so they moved with the field elevation of the aerodrome underneath them.

1.1 What changed

Change	Where
Nine A-CDM milestones derived from one classified **runway traversal**, replacing `ATOT/ALDT`	Line-up … Runway crossings
`airborne`, `touchdown` and `landing` are **interpolated instants**, not window extremes	Airborne, Touchdown, Landing
Parallel runways separated by **cross-track distance to each centreline**	Touchdown
The vertical profile follows **PRU's published methodology** — 200 NM radius, relocating tops, rolling-window level flight	Top of climb and top of descent, Level segments
The airport-layout gate and the level-segment floors are measured **above field elevation**	Airport surface events, Level segments
`AOBT/AIBT` renamed `off-block/on-block`; one detector, A-CDM vocabulary	Off-block and on-block
Scoring is **per aerodrome**, with a suppression floor and a truth-resolution column	Annex A — Accuracy by aerodrome

1.2 The ladder

Every comparison in this document is read off a **ladder**: eight configurations, applied cumulatively, run over the same trajectories against the same reference data. Rung names appear throughout the chapters below, so they are listed once here.

Rung	What it adds
`V00_v3_shipped`	The baseline: V3's shipped configuration (`events_v0.1.0`), reconstructed field by field. A plan check asserts it differs from what ships in exactly the eight fields the ladder moves — no more, so the baseline really is v0.1.0, and no fewer, so no behaviour ships without a rung measuring it.
`V01_runway_tiebreak`	Cross-track distance before bearing error when picking a runway. Carries **no configuration change** — a bug fix behind a flag is a bug you have promised to keep — so it shows no delta here, and its effect is read off the runway-identity table instead.
`V02_layout_agl`	The airport-layout gate measured above field elevation rather than against the 1013.25 datum.
`V03_pru_level`	The level-segment arm switched from ICAO's anchored band to PRU's rolling window.
`V04_level_geometry`	The level classification bound to the flight's aerodrome geometry, in two ways at once: the 3,000 ft and 1,800 ft floors measured above the field, **and** PRU's 200 NM analysis radius actually applied.
`V05_pru_tops`	The PRU tops published — **and** the level-off classification re-anchored onto them.
`V06_runway_milestones`	The nine A-CDM runway milestones, which also gates off `ATOT/ALDT/entry-runway/exit-runway`.
`V07_shipped`	The version string. A plan check asserts this rung equals `EventConfig()` field for field, so the ladder's verdict is about the configuration that actually ships.

Two rungs carry more than one field, and both say so above rather than letting the name imply otherwise.

V04 is two fields and one idea. The floors bind the classification to the aerodrome's *elevation* and the radius binds it to the aerodrome's *distance*; splitting them would give two deltas that are not separately interpretable, because a segment the radius excludes is one the floor never got to judge. The radius needs a flag of its own because it cannot otherwise be switched off: the distances arrive from a join that half the step needs anyway, so a detector applying the bound "whenever the distances are present" applies it always — including at a baseline meant to reproduce a version that declared the radius and read it nowhere.

V05 is two things and is not attributable to one. Publishing the PRU tops and re-anchoring the level-off classification onto them are separate decisions that arrive in one rung, so its delta is the sum of a new event family and a changed classification, and no part of it can be assigned to either alone. They are bundled because the anchor has no meaning until the tops exist. A reader wanting the two apart would need a rung that emits the tops without consuming them, which this ladder does not have.

1.3 What the evidence says

Table 1.1

Milestone	Coverage	Bias	MAD	p90
`airborne`	76.59%	+22 s	22 s	58 s
`touchdown`	80.37%	-6 s	18 s	41 s
`off-block`	9.10%	+720 s	722 s	1083 s
`on-block`	7.85%	-301 s	307 s	575 s

1.3.1 What publishing both families is worth

The table above compares each detector against the milestone it was designed to replace. This one asks the question a consumer actually faces: **given that `events_v0.2.0` publishes both methods, how often is the instant available at all?** Each row pools the two detectors for that milestone, which is exactly what the published table contains.

Table 1.2

APDF milestone	Detected	Reachable	Pooled coverage	Best single detector, same sample	Pooling adds
ATOT	11,235	11,675	**96.2%**	89.8% (`ATOT`)	+6.5 pts
ALDT	11,186	11,657	**96.0%**	95.9% (`ALDT`)	+0.0 pts

Read against the same sample, the older pair is the stronger of the two on coverage. On these three days `ATOT` alone answers 89.8% of reachable departures and `ALDT` 95.9% of arrivals, against 76.6% and 80.4% for `airborne` and `touchdown`. The A-CDM family was designed to supersede them and on this evidence it does not: it sees the aircraft at 15 ft, and there are movements where nothing is received that low.

Pooling still pays, but only on departures, and by less than a first reading of this table suggests. The pooled figure gains **+6.5 points over `ATOT`** because the two methods fail on different flights — where reception is too thin for the 15 ft crossing, the 1,500 ft climb window still resolves the movement, and where that window's gates reject a trajectory the runway traversal sometimes still carries it. On arrivals the gain is **+0.04 points**: `ALDT` already answers almost everything reachable, and `touchdown` adds essentially nothing a consumer could not already get.

An earlier draft of this chapter claimed pooling answered "a third more departures than either alone". That compared a 2026 pooled figure against V3's *2025* single-detector coverage — different period, different denominator — and overstated the gain by a factor of five. The numbers above are all from one run on one sample. The correction is recorded here rather than quietly fixed, because the retention decision was taken partly on the strength of the wrong number, and it survives the right one only for departures.

So `ATOT`/`ALDT` are retained on the evidence: for departures because pooling measurably helps, and for arrivals because `ALDT` outperforms its intended replacement outright. See `ATOT/ALDT` are retained, not retired. The A-CDM family's case rests on **timing**, not coverage — see Airborne.

1.3.2 How a consumer should actually pool them

The pooled row above is **not a rule anybody can ship.** `align` keeps whichever of the two detections is nearer the reference, which requires already knowing the reference: it is an oracle, and its bias is a floor no published table can reproduce. The question a consumer faces is the reachable one — *take one detector, fall back to the other* — so both orders are scored here against the oracle that bounds them.

Table 1.3

Milestone	Rule	Coverage	Bias	MAD	within 10 s
ATOT	legacy first	96.23%	+25 s	25 s	30.93%
ATOT	A-CDM first	96.21%	+25 s	25 s	24.72%
ATOT	*oracle*	96.23%	+23 s	25 s	31.72%
ALDT	legacy first	95.96%	+5 s	15 s	41.07%
ALDT	A-CDM first	95.96%	-2 s	18 s	25.48%
ALDT	*oracle*	95.96%	+1 s	10 s	49.49%

Coverage does not depend on the order, and cannot. A movement is answered if *either* detector answered it, so both rules reach the same 96.2% of departures and 96.0% of arrivals. Choosing a leader is therefore purely an accuracy decision, with no coverage cost either way — which is the useful thing to know before making it.

Prefer the older detector, fall back to the A-CDM one. On departures the two rules tie on median bias, both at +25 s — the reference's minute-rounding dominates — so the discriminating column is the tight one: `ATOT`-first puts **30.9%** of departures within ten seconds against A-CDM-first's **24.7%**. On arrivals `ALDT`-first is better on spread (**MAD 15 s against 18 s**, within ten seconds **41.1% against 25.5%**) even though A-CDM-first has the smaller median bias (-2.4 s against +5.0 s). A median closer to zero with a wider spread is the worse instrument: it is right on average and less often right on the movement.

The gap to the oracle says how much a cleverer rule could win. On departures almost nothing — 31.7% within ten seconds against the rule's 30.9%, so a per-movement chooser would buy under a point. On arrivals it is real: 49.5% against 41.1%. Something decidable at publication time — reception density at the aerodrome, or whether the traversal carried a surface sample — might close part of that, and this study does not attempt it.

Three properties of this study bound every number in it, and they are stated here rather than in a footnote.

The twenty aerodromes are the best-covered twenty, so every figure is an upper bound on network performance rather than a typical case — see The study set: twenty aerodromes.

Most reference times are rounded to the minute. A bias under a minute at such an aerodrome cannot be distinguished from APDF's own quantisation, which is why Annex A carries a resolution column and why comparing two aerodromes without it invites the wrong conclusion.

The reference join has a ceiling. Coverage is a share of reachable movements, and the reachable share is itself under 100% — see Ground truth, and its ceiling.

2 How to read a milestone chapter

Every milestone chapter has the same five parts, so it can be skimmed or read in full:

What it is — in operational terms, with its A-CDM/ICAO milestone number where it has one. **How it is detected** — the question the algorithm answers, then the mechanism. **Parameters** — each with its value, unit and *source*: ICAO's published value, PRU's, `traffic`'s, OpenAP's, or ours. **Compared with reference data** — the APDF scores, or why none is possible. **Limitations** — what is known to be wrong.

Three conventions apply throughout.

Storage is SI; thresholds are aviation. Altitudes are metres and vertical rates metres per second on disk, because the OSN schema mirrors OpenSky's own. Every threshold in this document is named in aviation units and scaled at the point of comparison.

Almost every threshold is named in `EventConfig` (`src/opdi/config.py`). That is what makes the parameter tables below a description of the code rather than a description of someone's memory of it, and it is what would make a sweep possible. Five exceptions remain — the three dead bands, the roll's gap bound and the arrival lead, all in the runway family — and each is labelled *module constant* in the table it appears in, and listed again in Limitations.

One engine produces every interpolated instant. Flight-level crossings, ring crossings, `airborne`, `touchdown` and `landing` all pass through the same Schmitt-trigger-and-interpolation code. It is defined once, in Flight-level crossings, and the four chapters that use it before that point say so and link there. A reader working front to back may want to read that chapter's first two paragraphs early.

`EventConfig.legacy()` still reproduces the published `events_v0.0.2` exactly, and the V4 ladder's baseline rung reproduces what V3 shipped (`events_v0.1.0`), so every comparison below is a measurement of two configurations on the same trajectories rather than a recollection.

3 Ground truth, and its ceiling

Reference data is EUROCONTROL APDF. It is in movement form — there is no literal take-off-time column, and `SRC_PHASE` decides what a row carries — and it has **no aircraft address**, so it cannot be joined to ADS-B directly.

The route runs through the flight table on `ID` — in PRISME `IM_SAMAD_ID`, the "SAM ID". `eurocontrol::apdf_tidy()` renames it to `ID` on the APDF side and `eurocontrol::flights_tidy()` returns it under the same name on the flight side, so the join is an exact key match rather than a fuzzy match on callsign and registration. It is null on 2.83% of APDF rows on the June 2025 extract; the measured share for this period appears below, and that, together with the reach rate, caps every coverage figure in this document.

Table 3.1: `ID` is null on 3.02% of APDF rows over this period.

Scope	APDF movements	Reaching `icao24`	Share
study	248,219	233,947	**94.25%**
network	1,206,139	1,140,970	**94.60%**

Three traps in this join fail silently, and all three are closed in the loader rather than left to callers.

The reference carries the commercial flight number; ADS-B carries the ATC call-sign. They disagree on roughly a fifth of rows — `LH416` against `DLH416` — and joining on the wrong one discards that fifth while looking exactly like a detection failure.

Timestamps are tagged `Europe/Paris` despite being named `_UTC`. The instants are correct, so converting is right and stripping the zone is a two-hour error. Spark's session timezone is asserted rather than assumed, because the distributed builder does not set it.

ADS-B callsigns are space-padded to eight characters. An untrimmed join matches nothing and reports every method at 0.00% rather than erroring.

3.1 Resolution differs by milestone, and by aerodrome

Most reporting aerodromes round movement and block times to the whole minute: on the V3 sample, 64% of `MVT_TIME_UTC` and 77% of `BLOCK_TIME_UTC` values landed exactly on a minute, while ring crossings are second-resolution throughout. **A perfect detector would still show tens of seconds of apparent spread against a minute-rounded truth**, and the spread it shows is then a property of the reference, not of the algorithm.

This is measured per aerodrome rather than assumed, from the share of reference times with a non-zero seconds field, and carried into Annex A as a column. It is the difference between reading LSZH's bias as an algorithmic result and reading EGCC's as one.

Table 3.2: Subscript w marks an n-weighted mean of the per-aerodrome medians, which is an approximation: medians do not pool. The exact per-aerodrome values are in Annex A — Accuracy by aerodrome.

Milestone	Truth resolution	Aerodromes	n truth	$Bias_w$	MAD_w	$p90_w$
`AIBT`	second	8	3,045	-282 s	286 s	2391 s
`AIBT`	minute	20	8,612	-402 s	347 s	3656 s
`ALDT`	second	3	2,050	-7 s	7 s	17 s
`ALDT`	minute	20	9,607	+8 s	15 s	37 s
`AOBT`	second	6	2,647	+620 s	620 s	2858 s
`AOBT`	minute	20	9,027	+829 s	839 s	2818 s
`ATOT`	second	4	2,032	+0 s	5 s	10 s
`ATOT`	minute	20	9,643	+29 s	31 s	55 s

4 The study set: twenty aerodromes

The period is **5, 6 and 7 June 2026**, scored against `apdf_202606`. The study set is the top twenty of the tier-A ADS-B coverage ranking computed over **those same three days**:

> EBBR, LSZH, EICK, EFHK, LEIB, EDDS, LFLL, LHBP, LFPO, LPFR, ESSA, ENVA, LOWW, LKPR, EDDP, LPPT, EGGD, EGNT, EGCC, UGKO — in rank order, which is the order Annex A uses.

These are the aerodromes where reception is best, so every figure in this document is an upper bound rather than a typical case. That is a deliberate design choice and it has a cost worth naming: a detector limited by reception rather than by its algorithm — the block milestones, most obviously — will look better here than it will anywhere else, and a reader extrapolating a network-wide coverage figure from these tables will overshoot. The correct reading is "this is what the algorithm can do where the data is good", and the distance between that and the network figure is a reception question, not an algorithm question.

Selecting the study set on a ranking computed over the very days being scored is the second half of the same caveat. It is not circular — the ranking is built from reception statistics, not from milestone accuracy — but it does mean the twenty are chosen for the sample rather than for the year.

4.1 Why 2026 and not V3's period

V3 used June 2025. V4 could not, and the reason is one aerodrome.

UGKO (Kutaisi) has 868 movements in `apdf_202606` and zero in `apdf_202506`. It sits at rank 20 of the coverage ranking, so dropping it would have meant either a nineteen-aerodrome study or promoting rank 21 and having the study set no longer be "the top twenty". Neither is wrong, but both would have to be explained; moving the period costs nothing else, because nothing in V4 depends on V3's month.

The consequence is that **V4's numbers are not directly comparable with V3's** figure for figure: a different month, a different traffic sample, and a track table rebuilt under the A8 segmentation default. Where this document compares against V3 it does so through the ladder's own baseline rung, which runs V3's configuration on V4's tracks, rather than by quoting V3's published table.

4.2 The three-day sample against a month of reference data

The reference extract covers a month; the detection sample covers three days. The scorer restricts the truth to the benchmark days before scoring, because scoring three days of tracks against a month of movements divides every coverage figure by roughly ten and looks like a catastrophic detector rather than a mismatched denominator. This is stated because it is the kind of error that produces a plausible-looking table.

5 Timing a runway movement: two methods, four milestones

Four of this study's milestones name the two instants every runway movement has – the lift-off and the touchdown. `ATOT` and `ALDT` name them the way OPDI has since v0.1.0; `airborne` (T08) and `touchdown` (T17) name them the way this family introduces. They are not variants of one algorithm. They are two different methods that answer the same two questions from the same trajectory, and every difference in the numbers that follow traces back to the method. This chapter states both once, so the per-milestone chapters can be about the result rather than the mechanism.

5.1 Method A — the window extreme (`ATOT`, `ALDT`)

`detect_runway_movements` reads the part of a trajectory that is unambiguously a departure or an arrival and takes its endpoint. For each flight it keeps the samples that are

- within `runway_max_dist_nm` = **5 NM** of the aerodrome the flight list names,
- below `runway_max_height_ft` = **1 500 ft above the field**,
- faster than `runway_min_groundspeed_kt` = **30 kt**, and
- climbing faster than `runway_min_vert_rate_ftmin` = **257 ft/min** (a departure) or descending faster than that (an arrival).

What survives is the **initial climb** or the **final descent**. The runway is named by the median ground track of those samples against each runway bearing (within `runway_max_bearing_deg` = 10 deg, the aircraft's cross-track distance to the centreline breaking ties between parallels). The time is then simply the **earliest** surviving sample for a departure – a lift-off proxy – and the **latest** for an arrival – a touchdown proxy.

Its character follows from that last step. The time is the *extreme sample of a detection window*, so the reasoning went that it must be displaced from the true instant by construction – the earliest climb sample OPDI received is already climbing – and V3 measured **+19 s median** on `ATOT`, which this study reproduced at +25 s.

That reasoning does not survive the measurement. At the three aerodromes whose reference resolves to the second, `ATOT` lands at **+1.0 s (EDDS), +4.0 s (LSZH) and −6.0 s (LEIB)**. The +19 s and +25 s are almost entirely APDF's minute-rounding, exactly as they are for the interpolated crossing. The window extreme is displaced *in principle*; at 5 s sampling in the initial climb the displacement is a few seconds, not tens of them, and no table in this study measures it as large. But because the window sits in the climb and descent, hundreds to 1 500 ft up, it reads on signal that ordinary receivers see everywhere – so it needs neither surface reception nor an airport layout, and its coverage is high.

5.2 Method B — the interpolated crossing (`airborne`, `touchdown`)

`runway_ops` first establishes *what the aircraft was doing on this runway* – a **traversal** (see Line-up) – and then reads a single geometric instant from it. A traversal is the run of samples whose H3 resolution-12 cell falls in the runway grid of an aerodrome the flight names: the runway rectangle itself (occupancy), and for an arrival the approach corridor short of the threshold. It is classified departure, arrival or crossing from its alignment, its speed and its height at entry and exit.

The milestone is then **not a sample at all**. `airborne` is the instant the height above the field crosses `runway_airborne_height_ft` = **15 ft** going *up*; `touchdown` is the same crossing going *down*. A Schmitt trigger with a 5 ft dead band decides *whether* the crossing happened – so a rotation that settles briefly does not emit twice – and the instant is obtained by **linear interpolation between the two samples that straddle 15 ft**, which is a different pair of rows from the pair that confirmed it. That is the same threshold-crossing engine Flight-level crossings describes.

Its character was expected to be the opposite of Method A's: an interpolated crossing is not displaced by construction, so it should carry no method bias. Measured against a second-resolution reference it lands at **+6.4 s (EDDS), +7.2 s (LSZH), −1.5 s (LEIB)** – small, but *not* smaller than Method A's on the same aerodromes. The theoretical advantage is real and the measured one is not there to be found. But it can only be read where the aircraft was received *near the runway*, at 15 ft rather than at 1 500 ft, and only for a flight whose aerodrome the flight list resolved – so its coverage depends on surface reception and on the runway grid in a way Method A's does not.

5.3 The differences, in one place

	`ATOT` / `ALDT` (Method A)	`airborne` / `touchdown` (Method B)
Detector	`detect_runway_movements`	`runway_ops` traversals
Associates by	distance + height window, runway by median track	H3 runway-grid occupancy + along/cross-track geometry
Reads the aircraft at	the climb / descent, up to 1 500 ft AGL	the 15 ft AGL crossing, on and just off the runway
The time is	the **extreme sample** of the window	the **interpolated** 15 ft crossing
Method bias, second-resolution aerodromes	+1.0 / +4.0 / −6.0 s	+6.4 / +7.2 / −1.5 s
Depends on	flight-list aerodrome only	flight-list aerodrome **and** surface/low reception **and** the runway grid
Height datum	height above field elevation	height above field elevation, gated at 15 ft

The trade the two methods make is coverage against accuracy, and it is not symmetric. Method A buys coverage by reading the well-received climb, and pays for it with a systematic bias no larger sample averages away. Method B buys accuracy with an interpolated crossing, and pays for it by needing the aircraft near the ground – which is where ADS-B reception is thinnest. The whole question this study asks of these four milestones is whether Method B's accuracy can be had without surrendering Method A's coverage; the Airborne and Touchdown chapters report the answer against APDF.

One caveat governs how that answer can even be read. The bias above is measured against APDF's `ATOT`/`ALDT`, and at seventeen of the twenty aerodromes APDF reports those times only to the **whole minute** (see Ground truth, and its ceiling). A whole-minute truth cannot resolve a few-second method difference – it injects tens of seconds of apparent bias that is the reference's rounding, not the detector's error. Method B's ~±2 s is therefore visible only at the three aerodromes whose APDF times are to the second (`LSZH`, `LEIB`, `EDDS`); the per-aerodrome tables carry the resolution flag for exactly this reason, and a bias read without it is a bias read wrong.

6 Line-up

What it is. A-CDM **T06**: the moment the aircraft enters the runway to depart. Published as `line-up`.

How it is detected. The question this family answers first is not "when did the aircraft line up" but **"what was this aircraft doing on this runway"** — and until that is answered, none of the nine milestones can be placed.

6.1 The runway traversal

Nine milestones come out of one object, described here once and referred to by every runway chapter that follows.

A **traversal** is a maximal run of consecutive state vectors whose H3 resolution-12 cell falls inside a runway polygon of an aerodrome the flight list names for that flight. A gap longer than `airport_trace_gap_seconds` starts a new one, so an aircraft that sits on the runway across a coverage hole becomes two traversals rather than one implausible ten-minute occupancy.

Each traversal is then classified by three measurements over its own samples:

- **alignment** — the angle between the traversal's median ground track and the runway's bearing, folded to [0°, 90°] so a reciprocal landing counts as aligned;
- **speed** — the maximum groundspeed reached;
- **height** — the height at entry and at exit above the field elevation of **the traversal's own aerodrome**.

giving one of three classes:

Class	Conditions
`departure`	aligned within 30°, faster than 50 kt, enters below 15 ft AGL and leaves at or above it
`arrival`	aligned within 30°, faster than 50 kt, enters at or above 15 ft AGL and leaves below it
`crossing`	at least 45° off the bearing, never above 40 kt, over in 120 s or less, below 15 ft AGL at both ends

or **NULL, which means abstain**. The gap between 30° and 45° is deliberate: naming the wrong milestone corrupts a movement count, while naming none is an explicit null a consumer can see. A traversal that falls in the band produces no events at all.

Which field the heights are measured against. Every height in this family is measured against the elevation of the aerodrome whose runway the traversal is on. That sounds like a detail and is not. A flight has two aerodromes and the pipeline attaches both elevations, so there is a cheaper reading available: take the *smaller* of the two field-relative heights. That reading is correct where it is used — deciding whether a sample is close enough to the ground to be matched against an airport layout, or to be called GND by the phase classifier — because a track is on the ground at only one of its two aerodromes, cruise sits far above both, and the wrong end can only ever make the height larger. Permissive is the safe direction for a membership test.

It is the wrong direction for a *threshold*. Ibiza's field is 24 ft and Zurich's is 1,416 ft; measured against the higher of the two, a departure from Ibiza has a height around −1,400 ft for its entire roll and climb-out, never reaches 15 ft, and is classified as neither a departure nor an arrival nor a crossing. The detector abstains, and `line-up`, `take-off-roll` and `airborne` are absent for the whole flight with nothing in the output to say why. Symmetrically an arrival at the lower of a flight's two aerodromes loses `landing`, `touchdown` and `runway-vacated`, and a go-around there is suppressed as a landing. Where the permissive height *does* clear the threshold, it clears it late by the elevation difference — a systematic bias in one direction, at exactly the aerodrome pairs where it is largest. So the runway family measures against the traversal's own aerodrome, the go-around against the destination's (it is arrival-anchored by definition), and only the layout gate and the phase classifier keep the permissive reading, which is what they want.

Why one object rather than nine detectors. Line-up, take-off roll and airborne are not independent questions; they are three readings of one departure. Deriving them from a shared, explicitly classified traversal means they cannot disagree about which runway, which aerodrome, or even whether the movement was a departure. The classification travels with every event in `info` — `traversal_class`, `align_deg`, `max_gs_kt`, `rwy_ident`, `apt_icao` — so the decision is auditable from the published row rather than reconstructible only from the code.

Which runway. A traversal is matched against both directions of every runway at the aerodrome and one is chosen: nearest **cross-track distance** to the extended centreline first, then smallest *unfolded* bearing error, then the designator. The cross-track term is what separates parallel runways; the bearing term is what separates the two ends of one strip, because both ends share a centreline and both score identically on alignment. Cross-track distances are rounded to 0.001 NM — about two metres — before the comparison, so that the two ends of one strip tie exactly rather than being separated by floating-point noise, while parallel runways hundreds of metres apart stay separable.

6.2 Back to line-up

`line-up` is the **entry sample** of a `departure` traversal: the first state vector inside the runway polygon. It is a sampled instant, not an interpolated one, and it is honest about that — the geometric event it names is the aircraft crossing the runway edge, which is a polygon boundary the H3 rasterisation only resolves to about 25 m.

Attributes are taken at the entry *time* through an explicit ordering, never with `first`/`last` inside a `groupBy`, which take partition order.

Parameters.

Parameter	Value	Source	Governs
`airport trace gap seconds`	300 s	published	when a traversal is considered ended
`runway align max deg`	30°	ours	alignment tolerance for departure/arrival

Table 6.1

`line-up` events on the sample: **26,317**.

Parameter	Value	Source	Governs
`runway cross min deg`	45°	ours	minimum angle for a crossing; the gap to 30° is the abstention band
`runway roll speed kt`	50 kt	ours	rolling rather than taxiing
`runway taxi speed kt`	40 kt	ours	ceiling for a taxiing crossing
`runway crossing max seconds`	120 s	ours	longest traversal still readable as a crossing
`runway airborne height ft`	15 ft	ours	the height that separates on-deck from airborne
H3 resolution	12	`H3Config`	matching granularity

Compared with reference data. None is possible. APDF records a runway designator (`AP_C_RWY`) but **no entry or exit time for any surface feature**, so there is nothing to score a line-up instant against. What can be checked is the runway *identity*, and that is done through the milestones APDF does time — see Touchdown.

Limitations. Requires the aerodrome's OSM layout to exist, which is built for large and medium aerodromes only, and requires surface reception at the runway edge, which is the sparsest reception in the whole trajectory. A traversal in the 30°–45° abstention band produces no line-up. The instant is a sampled one, so it carries up to one sample interval of error by construction.

7 Take-off roll

What it is. A-CDM **T07**: the start of the take-off run. Published as `take-off-roll`.

How it is detected. The question is *when did the aircraft stop taxiing and start going*, and the difficulty is that a single fast sample is not an answer: a high-speed turn-off, or one noisy groundspeed value, produces one.

So the detector looks for a **run** of samples inside the departure traversal whose groundspeed reaches `runway_roll_speed_kt`, and confirms the run by requiring it to be held for at least `runway_roll_min_seconds`. **The confirmation and the timestamp are deliberately different rows**: the hold is what makes the run credible, but the roll started at the run's *first* fast sample, not at the sample that finished proving it. Stamping the confirming sample would put every T07 one hold-length late, always in the same direction — which is the exact defect this version indicts `ATOT` for, and a module cannot claim that improvement and then reintroduce it.

The hold is wall-clock, so it needs a continuity bound: a traversal may legally span a gap of up to 300 s, and without a bound two fast samples either side of a four-minute reception hole would satisfy a 5 s hold and stamp a roll nobody observed. A gap longer than `ROLL_MAX_GAP_SECONDS` therefore breaks the run.

Parameters.

Parameter	Value	Source	Governs
`runway roll speed kt`	50 kt	ours	rolling rather than taxiing; between taxi (~20 kt) and rotation (~140 kt)
`runway roll min seconds`	5 s	ours	how long the speed must be held before the run is credible
`ROLL MAX GAP SECONDS`	30 s	ours (module constant)	longest gap a roll may span — six times the nominal 5 s cadence

`ROLL_MAX_GAP_SECONDS` is a module constant rather than an `EventConfig` field, which is an inconsistency with the rest of this document and is recorded as one in Limitations. It is a tuning parameter and belongs in the config surface.

Compared with reference data. None is possible. APDF has no start-of-roll time. The nearest quantity it holds is ATOT, which is the *end* of the roll and is scored in Airborne.

Table 7.1

`take-off-roll` events on the sample: **26,008**.

Limitations. Depends on groundspeed being present and plausible on the runway, which is where surface reception is worst. A departure that is first received already above 50 kt has its roll stamped at the first sample received, not at the real start, and nothing in the output distinguishes the two. The 5 s hold is a floor, not a model of acceleration.

8 Airborne

What it is. A-CDM **T08**: wheels off the ground. Published as `airborne`. This is the milestone `ATOT` was trying to be. It is published *alongside* `ATOT`, not in place of it — see `ATOT/ALDT` are retained, not retired.

How it is detected. The question is *when did the height above the field pass fifteen feet*, and the answer must not be "at the first sample that was above it" — that is the whole defect being fixed here.

The height above field elevation is run through the same threshold-crossing engine as the flight-level and ring crossings, described in Flight-level crossings: a Schmitt trigger with a five-foot dead band decides *whether* the crossing happened, and the instant is then obtained by **linear interpolation between the two samples that straddle 15 ft**. That is a different pair of rows from the pair that confirmed it.

The contrast with `ATOT` was meant to be the point of the whole family. `ATOT` is the *earliest surviving sample* of a detection window bounded by distance, height, speed and vertical rate, and an extreme of a window is displaced by construction, which V3 measured at +19 s median on departures.

The V4 measurement does not support that contrast. Against a second-resolution reference `ATOT` is +1.0 s at EDDS and +4.0 s at LSZH, against `airborne`'s +6.4 s and +7.2 s; only at LEIB is `airborne` the better of the two (−1.5 s against −6.0 s). The large pooled biases on both sides are the reference's own minute-rounding. What `airborne` does win, and by a wide margin where it wins at all, is *coverage in the places `ATOT` fails* – see [Per-detector accuracy by aerodrome].

Parameters.

Parameter	Value	Source	Governs
`runway airborne height ft`	15 ft	ours	the height taken as wheels-off
`AIRBORNE HYSTERESIS FT`	5 ft	ours (module constant)	dead band, so a rotation that settles briefly does not emit a second `airborne`
plus the traversal parameters	—	—	see Line-up

Fifteen feet rather than zero because barometric altitude at the field is noisy at the foot level and a zero crossing would chatter. It is a stated proxy for wheels-off, not a claim to have measured the wheels.

Compared with reference data. Scored against APDF's ATOT.

Table 8.1

Detector	Rung	n truth	Coverage	Bias	MAD	p90	within 60 s
`ATOT`	V07_shipped	11,675	89.77%	+25 s	25 s	60 s	83.14%
`airborne`	V07_shipped	11,675	76.59%	+22 s	22 s	58 s	69.97%

Both rows are the same rung, and that is the point. An earlier draft read `ATOT` at `V00_v3_shipped` and `airborne` at `V07_shipped`, because the two were expected never to coexist – the flag that emitted one was to gate off the other. `events_v0.2.0` retains both (see `ATOT/ALDT` are retained, not retired), so the comparison no longer has to cross rungs: same run, same trajectories, same reference, two definitions of the same instant. That also removes the trap in the cross-rung form, which was to invite comparison against whatever *other* sample the baseline rung happened to be measured on.

Table 8.2

Rung	Coverage	Bias	MAD	p90
V06_runway_milestones	76.59%	+22 s	22 s	58 s
V07_shipped	76.59%	+22 s	22 s	58 s

Limitations. Fifteen feet AGL is a proxy, and its residual bias is reported rather than assumed away. The interpolation is linear between two samples up to one interval apart, so a rotation inside a 5 s gap is interpolated across a manoeuvre that is not linear. Requires a departure traversal, so it inherits every limitation of Line-up: OSM layout coverage, surface reception, and the abstention band.

9 Landing

What it is. A-CDM **T16**: the aircraft crosses the landing threshold. This is the milestone the type string `landing` now carries, and **it is not what `landing` meant under `events_v0.0.2` or `events_v0.1.0`** — see Breaking changes for consumers, which leads with exactly this.

How it is detected. The question is *when did the aircraft pass over the threshold*, which is a geometric question about position, not about height or speed.

Along-track distance from the threshold is computed as

$$d = \text{haversine}(\text{thr}, P) \cdot \cos(\text{bearing}(\text{thr}, P) - \text{rwy bearing})$$

which is negative short of the threshold and positive beyond it. The crossing is therefore a **sign change**, and it interpolates linearly in exactly the way the height crossings do — through the same engine, with a 0.05 NM dead band on confirmation and the instant taken from the sample pair that straddles zero.

One gate is applied: only crossings **above `runway_airborne_height_ft`** are kept. An aircraft rolling past the threshold on the ground — backtracking, or vacating via a turn-off beyond it — has already landed, and stamping a second `landing` there would double the arrival count.

T16 needs a sample the traversal does not contain. A sign change needs a point on each side of it, and the negative side is *short of the threshold* — which is outside the runway polygon, because the threshold is the polygon's near edge. So the samples belonging to an arrival traversal, which are by definition the samples inside the polygon, contain no negative along-track distance at all and the crossing can never be confirmed from them. For arrival traversals only, the sample window is therefore extended **backwards by `ARRIVAL_LEAD_SECONDS` (60 s)** before the traversal's entry, admitting the final-approach samples the crossing interpolates from. Nothing else reads the extended window: the entry and exit instants, the reported positions and every other milestone in this family stay bounded by the polygon, because those are statements about occupancy and the lead is not.

Sixty seconds is about 2.3 NM at approach speed — comfortably more than any displaced threshold or the gap between the polygon edge and the last approach sample, and far short of a previous traversal the same aircraft could have made at the same aerodrome. It is a property of the geometry rather than a tuning choice, which is why it is a module constant; the caveat in Limitations about module constants applies to it too.

Parameters.

Parameter	Value	Source	Governs
`runway airborne height ft`	15 ft	ours	floor below which a threshold crossing is ground movement, not a landing

Parameter	Value	Source	Governs
`THRESHOLD PLANE HYSTERESIS NM`	0.05 NM	ours (module constant)	dead band on confirmation; ~304 ft, about 1.3 s at approach speed
`ARRIVAL LEAD SECONDS`	60 s	ours (module constant)	how far before the traversal's entry the arrival sample window reaches, so the pre-threshold sample exists
plus the traversal parameters	—	—	see Line-up

The dead band's width does not decide the reported instant, which is worth stating because it is easy to assume otherwise: the band gates *confirmation*, and the instant is interpolated to the bare plane, earlier than the confirming sample and unaffected by how wide the band is.

Compared with reference data. None is possible for the instant. APDF's `MVT_TIME_UTC` for an arrival is ALDT — the touchdown — not the threshold crossing, and the two differ by the flare and the distance from threshold to touchdown point. Scoring T16 against ALDT would measure that offset and call it an error, so the shipped configuration deliberately **does not** map `landing` onto ALDT; Touchdown carries that comparison instead. The benchmark enforces this by construction rather than trusting it: the type-to-milestone map is narrowed per rung, so `landing` is simply absent from it on any rung with the A-CDM family switched on, and a plan check refuses a map that widens.

Limitations. The threshold position comes from OurAirports, so a runway whose thresholds are recorded imprecisely displaces every T16 on it by the same amount — a per-runway bias with nothing in the output to show it. The along-track projection assumes the aircraft is approaching roughly along the centreline, which is true of a stabilised approach and less true of a circling one. And it inherits the runway-direction pick from Line-up, whose known failure mode is described in Cross-cutting findings: on a runway near bearing 360 the median track can resolve to the reciprocal, and the landing is then measured from the wrong end of the strip. Finally, an approach whose last reception before the threshold is more than 60 s old falls outside `ARRIVAL_LEAD_SECONDS` and yields no T16 — the aircraft still gets its `touchdown` and `runway-vacated`, so this shows as a `landing` count below the `touchdown` count rather than as a lost arrival.

Table 9.1

`landing` (T16) events on the sample: **4,004**.

10 Touchdown

What it is. A-CDM **T17**: the wheels meet the runway. Published as `touchdown`, published *alongside* `ALDT` rather than in place of it (see `ATOT/ALDT` are retained, not retired).

How it is detected. The mirror of Airborne: the same height-above-field signal through the same crossing engine, in the **downward** direction. The instant is the interpolated crossing of `runway_airborne_height_ft`, taken from the pair of samples that straddle it.

`ALDT` was the *latest surviving sample* of the arrival detection window — the same window-extreme construction as `ATOT`, with the vertical-rate gate inverted.

Parameters. Identical to Airborne; the traversal's class decides the direction of the crossing.

Compared with reference data. Scored against APDF's ALDT. As in Airborne, the table is cross-rung by necessity: `ALDT` is read at `V00_v3_shipped` and `touchdown` at `V07_shipped`, because the flag that emits one gates off the other.

Table 10.1

Detector	Rung	n truth	Coverage	Bias	MAD	p90	within 60 s
`ALDT`	V07_shipped	11,657	95.92%	+5 s	15 s	45 s	94.58%
`touchdown`	V07_shipped	11,657	80.37%	-6 s	18 s	41 s	79.21%

Table 10.2

Rung	Coverage	Bias	MAD	p90
V06_runway_milestones	80.37%	-6 s	18 s	41 s
V07_shipped	80.37%	-6 s	18 s	41 s

10.1 Runway identity

This is where the runway the traversal named is checked, against `AP_C_RWY`. V3 measured a third of named landing runways wrong, and traced it to the tie-break: parallel runways were separated by **the distance from each threshold to the aerodrome reference point**, which is a constant of the airport, identical for every flight, and therefore picked the same runway of a parallel pair every time. V4 separates them by the aircraft's own cross-track distance to each extended centreline.

Table 10.3

Detector	Milestone	Named	Exactly right	Share of named
`touchdown`	ALDT	9,369 (80.37%)	8,672	**92.56%**

<table>
<tr><th>Detector</th><th>Milestone</th><th>Named</th><th>Exactly right</th><th>Share of named</th></tr>
<tr><td>| AOBT | 0 (0.00%) | 0 | **-** | | `airborne` | ATOT | 8,945 (76.62%) | 7,698 | **86.06%** | |</td><td>AIBT</td><td>0 (0.00%)</td><td>0</td><td>—</td></tr>
</table>

The fix is **not behind a configuration flag**, which is why the V4 ladder's `V01_runway_tiebreak` rung carries no configuration change and shows no delta: a bug fix behind a flag is a bug you have promised to keep. Its effect is read off the runway-identity table against V3's published equivalent, not off a rung, and the ladder exempts that rung by name so a *different* accidental no-op still fails the plan check.

Designators are compared case-insensitively with whitespace stripped and are not otherwise fuzzed, because `07R` and `07L` are different runways and a lenient match would hide exactly the error worth finding.

Limitations. Fifteen feet AGL is a proxy for the wheels, as in Airborne. Requires an arrival traversal. And the identity comparison is only as good as `AP_C_RWY`, which is itself a reported field.

11 Runway vacated

What it is. A-CDM **T19**: the aircraft leaves the runway. Published as `runway-vacated`.

How it is detected. The **exit sample** of an `arrival` traversal — the last state vector whose H3 cell is inside the runway polygon. Sampled, not interpolated, for the same reason as Line-up: the geometric event is a polygon boundary crossing, and the polygon is rasterised.

Attributes are taken at the exit *time*, through an explicit ordering.

Parameters. The traversal parameters, see Line-up. No parameter is specific to this milestone.

Compared with reference data. None is possible. APDF has no runway-vacated time; the next arrival milestone it records is AIBT, on the stand.

Limitations. Runway occupancy time — T17 to T19 — is the derived quantity most consumers will want from this pair, and it is bounded below by the rasterisation: a turn-off inside a single 5 s sample is not resolved. A traversal that ends because reception ended, rather than because the aircraft vacated, produces a `runway-vacated` at the wrong instant with nothing marking it.

Table 11.1

`runway-vacated` events on the sample: **36,449**.

12 Go-around

What it is. An approach abandoned and flown away from. It has **no A-CDM milestone number**, and is absent from the numbering rather than assigned an invented one.

How it is detected. The question is *did the aircraft descend towards this runway, not land, and climb away* — and the important design decision is that **it does not use a runway traversal at all**. An approach abandoned at 400 ft never overflies the strip, so a detector hung off a runway polygon would miss precisely the go-arounds that matter.

Instead the height above field elevation is run through the crossing engine at two heights: a **trigger** at 500 ft, which the aircraft must descend below, and a **recovery** at 1,500 ft, which it must regain. Each recovery is paired with the most recent descent below the trigger, and only the first recovery of an excursion is kept, so one excursion is one event. The samples inside that window are then gated three ways:

- the lowest point of the excursion stayed **above** `runway_airborne_height_ft` — if it did not, the aircraft touched down, however briefly, and the climb away is a departure, not an abandoned approach;
- the maximum rate of climb reached `goaround_min_roc_ftmin` — a climb away, not a descent that merely stopped descending;
- the low point was within `goaround_radius_nm` of the arrival aerodrome — the excursion happened on final, not somewhere else entirely.

The event is stamped at the low point, and its `info` carries `low_height_ft`, `recovery_roc_ft_min` and `excursion_seconds` so a consumer can see the shape of the excursion behind the label.

Defining "no touchdown in between" as *the excursion never reached 15 ft AGL* rather than as *no `touchdown` event falls inside the window* is deliberate: 15 ft is the same threshold that **defines** Touchdown, so the two cannot disagree, and the go-around detector keeps its independence from the traversal machinery.

Parameters.

Parameter	Value	Source	Governs
`goaround trigger height ft`	500 ft	ours	how low the approach must get to count as one
`goaround recovery height ft`	1,500 ft	ours	how high the climb-away must reach
`goaround min roc ftmin`	500 ft/min	ours	separates a climb away from a stopped descent
`goaround radius nm`	10 NM	ours	how close to the aerodrome the excursion must happen

Parameter	Value	Source	Governs
`GOAROUND HYSTERESIS FT`	50 ft	ours (module constant)	dead band on both heights, so a level-off at 500 ft does not toggle repeatedly
`runway airborne height ft`	15 ft	ours	the touchdown suppression

Compared with reference data. None is possible. APDF records one movement per flight; a go-around leaves no row of its own, and there is no EUROCONTROL-side count of abandoned approaches for OPDI to score against. The only external check available is a plausibility one — the rate, against the published rates of roughly one to three per thousand approaches — and a rate that lands in that band is consistent with the detector being right, not evidence that it is.

Limitations. Unvalidated. The four thresholds are ours, chosen to describe the manoeuvre rather than fitted to anything. A missed approach flown as a published procedure that levels at 3,000 ft is caught; a low-energy go-around that never descends below 500 ft is not. The detector needs the arrival aerodrome from the flight list for its radius gate, so a flight with no ADES produces none. And the low point is chosen by a minimum that can tie between two samples at the same height, which makes the timestamp non-deterministic by one sample interval on a flat minimum.

Table 12.1

`go-around` events on the sample: **440**.

13 Runway crossings

What it is. An aircraft taxiing across a runway rather than using it. Published as `runway-crossing-entry` and `runway-crossing-vacated`. **No A-CDM number**, and deliberately so: a crossing is not a movement.

How it is detected. The `crossing` class of the traversal classification in Line-up — at least 45° off the runway bearing, never above 40 kt, over within 120 s, and below 15 ft AGL at both ends. The two events are the traversal's entry and exit samples.

They exist to be separable from movements, which is what `entry-runway/exit-runway` were not. The published pair fired for a departure, an arrival and a taxi crossing alike, so any consumer counting runway occupancies was counting three different things under one name. Splitting them means a crossing gets its two instants and nothing else, and can never be added to a movement count by accident.

Parameters.

Parameter	Value	Source	Governs
`runway cross min deg`	45°	ours	minimum angle off the bearing
`runway taxi speed kt`	40 kt	ours	speed ceiling
`runway crossing max seconds`	120 s	ours	duration ceiling; a hold on the runway is not a crossing
`runway airborne height ft`	15 ft	ours	both ends must be on the deck

Compared with reference data. None is possible. No source available to OPDI records taxi crossings of a runway.

Limitations. The runway *direction* named on a crossing is weakly determined and a consumer should not read it. Cross-track distance identifies the strip correctly, but for a perpendicular traversal the bearing error is near 90° for both ends, so the choice between them falls through to the designator — i.e. to the alphabet. It is right about the strip and arbitrary about the direction, which is defensible for a manoeuvre that has no direction, but counting crossings per runway *direction* would be reading noise.

Table 13.1

`runway-crossing-entry` **89**, `runway-crossing-vacated` **89**.

14 Off-block and on-block

What it is. A-CDM **T04** (off-block) and **T21** (on-block) — the inputs to ICAO's KPI02 and KPI13. Published as `off-block` and `on-block`.

These are the events V3 published as `AOBT` and `AIBT`. One detector, two names: the A-CDM vocabulary flag renames them, and nothing about the detection changed. The rename is listed in Breaking changes for consumers.

How it is detected. The question is *when did the aircraft start moving*, and the trap is that a groundspeed field jitters: at 5 s sampling, a stationary aircraft produces occasional non-zero speeds.

So movement is defined as **sustained**: groundspeed above `ground_speed_threshold_kt` held for at least `ground_move_min_seconds`. The start of the first such run is the off-block candidate and the end of the last is the on-block candidate. This is `traffic`'s `StartMoving` signal, ported to column expressions.

Both are then **anchored on the parking-position events** from Airport surface events: a candidate is only emitted for a track actually seen leaving or entering a stand. Without that anchor, a track first received mid-taxi would report its first movement as an off-block — a different event, and one that reads as an implausibly short taxi rather than as a miss.

`traffic`'s own parking-position path is deliberately not used: it resolves stands through a live OpenStreetMap Overpass query whose cache does not survive serialisation to an executor.

Parameters.

Parameter	Value	Source	Governs
`ground speed threshold kt`	2 kt	`traffic`	what counts as moving
`ground move min seconds`	30 s	`traffic`	how long movement must persist

Compared with reference data. Scored against APDF's AOBT and AIBT.

Table 14.1

Detector	Rung	n truth	Coverage	Bias	MAD	p90	within 60 s
`off-block`	V07_shipped	11,674	9.10%	+720 s	722 s	1083 s	0.01%
`on-block`	V07_shipped	11,657	7.85%	-301 s	307 s	575 s	0.02%

All four rows are the same rung, and that is the point — the same correction made in Airborne. `events_v0.2.0` retains `AOBT`/`AIBT` alongside `off-block`/`on-block`, so nothing forces this comparison across rungs any more; all four are read at `V07_shipped`, and the pairs agree because they are the same detector under two names, not because two runs happened to agree.

14.0.1 What geometry and the ground-admission gate were each worth

This chapter's own plan predicted that coverage here was a reception limit, and that giving every study aerodrome stand geometry would lift it substantially. Measured, that prediction does not hold. Two fixes landed ahead of this measurement, and they are not the same size.

The published `hexaero_airport_layouts` table carried stand geometry for only fifteen aerodromes before this study, against twenty in the study set — and, separately from coverage, an aerodrome with no `parking_position` cells at all cannot anchor either milestone, since the detector requires a parking-position sighting on the way in or out, per Airport surface events. Two aerodromes, ENVA and EGNT, recorded exactly zero AOBT and AIBT detections both before this rebuild and after it — evidence that whatever geometry they now carry is still not producing an anchor there, at least on this sample. Which of the remaining eighteen genuinely had no stand geometry in the old table, as against having some that was too sparse to matter, is not separately known from this measurement; what is known is the per-aerodrome result below, which is not the broad lift either explanation alone predicts. Separately, `airport_admit_on_ground` — the fix that stops in-stand ADS-B surface reports from being discarded at an altitude gate that ADS-B ground messages cannot satisfy in the first place — had already landed before this measurement. Its effect is visible in the *starting point*: pooled coverage going into this rebuild was already 7.30% (AOBT) and 6.58% (AIBT), against a ~1% campaign baseline at nineteen of twenty aerodromes beforehand. **That gate fix is worth roughly six points of the twenty-point gap between "~1%" and where this family now stands — the larger of the two fixes, and the one this chapter previously did not credit at all.**

The geometry rebuild measured here — the published table growing to 1,036 aerodromes, all twenty study aerodromes now carrying parking-position cells — adds a further 9.10% pooled AOBT coverage and 7.85% AIBT, up from 7.30%/6.58%: **roughly two points, not the step change the plan predicted.**

Table 14.2

Aerodrome	AOBT before	AOBT after	AIBT before	AIBT after
EDDS	1.63%	42.51%	1.05%	23.47%
LKPR	1.66%	4.84%	1.95%	3.45%
UGKO	0.00%	4.55%	0.00%	7.50%
EBBR	1.18%	1.18%	1.58%	1.70%

Almost all of that two-point gain is one aerodrome. EDDS AOBT coverage moves from 1.63% to 42.51% (156 detections against 6 before) and AIBT from 1.05% to 23.47%. EDDS alone accounts for 150 of the 210 additional AOBT detections pooled across all twenty aerodromes. LKPR moves from 1.66%/1.95% to 4.84%/3.45%, and UGKO — which had zero of either milestone before, at any coverage — reaches 4.55% AOBT and 7.50% AIBT, both on a handful of detections. **Every other aerodrome in the study moved by a point or less,** including several that gained stand geometry in this rebuild for the first time. Two, ENVA and EGNT, are still at exactly 0.00% on both milestones.

EBBR is a case worth naming rather than leaving to be misread. This rebuild's OSM extract assigns one apron feature — 445 H3 cells — to EBMB (Brussels Melsbroek, the co-located military aerodrome with its own `aeroway=aerodrome` polygon nested inside EBBR's) rather than to EBBR, where Overpass's geocoded place polygon had previously over-reached and claimed it. Since `calculate_airport_events` joins on `array_contains(sv.apt, hexaero_apt_icao)`, an EBBR-labelled flight parked on that apron now produces no off-block or on-block event at EBBR at all — correct modelling of two distinct aerodromes, but a real change to what EBBR's geometry contains. **It did not move EBBR's measured coverage**: AOBT stayed at exactly 1.18% (10 detections, both before and after), AIBT moved from 1.58% to 1.70% — because EBBR was already at the reception floor with the rest of the non-EDDS aerodromes, not because the apron reassignment pushed it there. A reader comparing EBBR's Annex A row against an earlier version of this table should attribute the difference to that floor, not to the EBMB split.

So the residual is reception, but it is a residual, not the whole shortfall. Fifteen points of the twenty-point rise from ~1% to where this family now stands came from a gate that was discarding ADS-B ground reports before they could be scored at all; roughly two more points came from geometry, concentrated almost entirely at one aerodrome large enough and well-enough OSM-mapped for the fix to matter; what neither fix touches is covered next, and it is the more important finding of this measurement.

14.0.2 The timing is the real defect, and it is not a reception problem

Both milestones carry a large, one-sided, and essentially uniform bias. Pooled AOBT fires +720 s (MAD 722 s) after the reference AOBT — about twelve minutes late — and pooled AIBT fires -301 s (MAD 307 s) *before* the reference AIBT — about five minutes early. `mad_s` tracks `|bias_s|` closely at every aerodrome this scores, which is the signature of a systematic offset, not scatter: within-60-second accuracy is at or near 0.00% almost everywhere, including at LSZH and EDDS where coverage is highest. The bias is present at every aerodrome regardless of how well it is covered — +722 s at LSZH, +705 s at EDDS, +765 s at LKPR — which rules out reception density as the cause: a reception problem would be worst where coverage is worst, and this is not that.

The same run, the same aerodromes, contradicts a reception explanation directly. At LSZH in this measurement, `ATOT` lands at +4 s with 96.04% of departures within ten seconds, and `ALDT` at -5 s. Both read ADS-B surface and near-surface data at the same aerodrome, the same period, the same reception conditions as the block-time detector. If reception at the stand were the limiting factor, the runway milestones next to it would show the same order-of-magnitude bias. They do not.

A hypothesis, stated as one, not as a finding: the anchor is probably the wrong cell, not a missing one. `calculate_block_events` anchors off-block on the *last* parking-position cell a track touches before it starts moving, and on-block on the *first* one it touches after. At an aerodrome with stands lining the taxiways, an aircraft pushing back and taxiing out keeps crossing other aircrafts' stand polygons for minutes after it has left its own — so the *last* cell it touches is well into the taxi, which reads as a late off-block. Taxiing in, the *first* stand-shaped cell it crosses is upstream of its own gate, which reads as an early on-block. That predicts exactly the observed asymmetry — late departure, early arrival — at a magnitude set by taxi distance rather than by reception. The fix, if this holds up, would be to anchor on the stand where the aircraft actually **dwelled**, not the first or last cell brushed in passing. That is out of scope for this measurement and needs its own re-run to test; it is recorded here as the most likely next step, not as something this study established.

Limitations. Not reception-bound at the stand, or not only — that was this chapter's account until this measurement, and it is now only the residual after two identified fixes: an altitude gate worth roughly six points of coverage and a stand-geometry rebuild worth roughly two more, concentrated almost entirely at one aerodrome. **What neither fix touches, and what makes these unusable for KPI02 or KPI13 regardless of coverage, is a systematic**

bias of about twelve minutes on off-block and five minutes on on-block, present at every aerodrome including the best-covered ones, and consistent with the detector anchoring on the wrong stand cell rather than with any reception limit. Pushback (T05) was not built for the same reason V3 gave and this measurement does not change: it needs strictly more coverage than off-block has, and off-block's own accuracy is not yet usable to build on.

15 Top of climb and top of descent

What it is. The end of the climb and the start of the descent. Two algorithms are published side by side, under four type names:

Type	Algorithm	`info.method`
`top-of-climb` / `top-of-descent`	the fuzzy phase classifier	`phase`
`top-of-climb-cco` / `top-of-descent-cdo`	EUROCONTROL PRU's vertical-efficiency methodology	`pru`

Both are published because neither can be shown better. No reference data records a top of climb — APDF holds movement and block times, ring crossings, a runway and a stand, and nothing about the cruise phase. Publishing one and suppressing the other would be a preference presented as a finding. Every top now names the algorithm that produced it, so a consumer can choose.

15.1 How the phase tops are detected

Directly from the phase labels: the earliest and latest sample classified cruise. The cruise rule is `min(alt_hi, roc_zero, spd_hi)` with OpenAP's membership functions — altitude Gaussian at 35,000 ft (20,000), vertical rate Gaussian at 0 ft/min (100), speed Gaussian at **600 kt (100)**.

Those are OpenAP's published values, not chosen here, and they carry a known consequence: a turboprop cruising at 250 kt scores `exp(-6.125)`   `0.002`, so the cruise rule never wins and **turboprop and general-aviation traffic get no phase top of climb or descent at all**. It is inherited rather than an OPDI defect, and left in place deliberately: deviating from a reference constant should follow a measurement, not precede one.

15.2 How the PRU tops are detected

The question PRU asks is different, and the difference is not a tuning choice. Its methodology is published at https://ansperformance.eu/methodology/cd-vertical-flight-efficiency-pi/; the quantities below are transcribed from it and from the PRC technical note.

The analysis is bounded by a 200 NM radius around the flight's own aerodromes. `D200` is the first crossing of that radius outbound, `A200` the last inbound. A step climb 250 NM out is cruise-altitude optimisation, not a top of climb, and the radius exists precisely to exclude it.

ToC-D200 is the first 4D point of highest altitude between take-off and D200 inclusive; **ToD-A200** is the last 4D point at which the aircraft leaves the highest altitude between A200 and touchdown. First and last respectively, not either end of a plateau — a climb is interrupted from below and a descent from above.

Then the top moves. From the ToC-D200 altitude down to **90%** of it is the *exclusion box*. A level segment inside that box lasting over **5 minutes** is cruise rather than a level-off: it is dropped from the level-off family, and the top relocates — **ToC-CCO** to the start of the first such segment, **ToD-CDO** to the end of the last. With no such segment, ToC-CCO is the ToC-D200 top, and `info` records `toc_relocated` so the two cases are distinguishable in the published data.

A track that never reaches the 200 NM radius keeps its whole trajectory in the analysis window — the fallback is the last sample outbound and the first inbound. A short sector is a sector PRU still has tops for, and abstaining would drop every flight under about 400 NM, which is most of them.

Parameters.

Parameter	Value	Source	Governs
`level analysis radius nm`	200 NM	PRU	the ring bounding the analysis
`level exclusion box pct`	90%	PRU	lower edge of the exclusion box
`level exclusion box seconds`	300 s	PRU	duration above which a segment in the box is cruise
`emit pru tops`	on	ours	whether the PRU pair is emitted alongside the phase pair
`level anchor`	`pru`	ours	which pair anchors the level-off classification
`phase cruise speed kt`	600 kt	OpenAP	centre of the phase cruise-speed membership
`phase cruise speed sigma kt`	100 kt	OpenAP	its width

Table 15.1

Type	Events
`top-of-climb`	97,229
`top-of-descent`	96,870
`top-of-climb-cco`	172,533
`top-of-descent-cdo`	172,515

The count difference between the two families is itself informative: the PRU tops do not require a cruise phase to be recognised, so they exist for the turboprop traffic the phase tops silently omit.

Compared with reference data. None is possible, for either algorithm, for the reason given above. The claim the PRU pair makes is **conformance to a published methodology**, tested on synthetic trajectories whose geometry is known by construction: a climb to 30,000 ft with a 360 s hold at 27,500 ft (91.7% of the top, inside the box, longer than 300 s) must relocate the top; a 240 s hold at the same altitude must not; a 600 s hold at 20,000 ft (67%, below the box) must not, and must be reported as a level-off instead. Those three cases are the whole rule and each is a test.

Limitations. The phase pair inherits the cruise-speed bias described above. The PRU pair depends on the flight list naming both aerodromes: without a position for the aerodrome there is no distance, and a NULL distance fails the radius test rather than passing it — a segment that cannot be placed relative to an aerodrome is outside the methodology, not inside it by default. Neither pair is validated against data, because there is nothing to validate against.

16 Level segments

What it is. Level flight during climb and descent — the inputs to ICAO's KPI17 and KPI19 — published as `level-off-climb-start` / `-end` and `level-off-descent-start` / `-end`.

Separately, and **not interchangeably**, OPDI continues to publish `level-start` and `level-end` from the fuzzy phase classifier. Those ask whether a sample *looks like* level flight; the level-off family asks a geometric question about altitudes. Both are published, under different names, and a consumer computing a vertical-efficiency indicator wants the latter.

16.1 How it is detected

Two arms are implemented and one is selected by `level_method`. The shipped default is PRU's.

The PRU arm (`pru`, shipped). The question is *did the aircraft stay level*, answered as a property of the altitudes themselves rather than as a test on the reported vertical rate — which matters because `vert_rate` is a broadcast field with its own noise and its own staleness. A trajectory part is level when it stays inside a rolling window whose dimensions satisfy **Y/X = 300 ft/min**: at the default 10 s window, a 50 ft box. The trajectory is linearly interpolated onto that grid, because a discrete track has no sample at every instant the window requires. A segment is counted from **20 s** long.

The window height is *derived*, not stored: `level_window_height_ft` is `level_vertical_speed_limit_ftmin × level_window_seconds / 60`. Storing both dimensions would let them drift, and a window whose ratio is not 300 ft/min is not the PRU window under a different parameterisation — it is a different definition of level flight wearing the same name.

The ICAO arm (`icao`, retained). The algorithm shipped in `events_v0.1.0`, kept reachable and unchanged. ICAO specifies it completely:

> a data point starts a level segment when the altitude difference with the next data point is the level band limit and the vertical speed towards it is the vertical speed limit. The segment ends when the altitude differs from the segment's starting altitude by more than the level band limit, or the vertical speed between two consecutive points exceeds the limit.

The anchoring is what a plain window function cannot express: a segment's end is defined against *its own start*, a running value that resets. It is implemented as a sessionisation — mark where flatness begins, forward-fill the anchor within the group — with each group truncated at its first failure, so a trajectory that leaves the band and drifts back is not stitched across the excursion.

Classification, common to both arms. Segments are split into KPI17 and KPI19 by whether they fall before the top of climb or after the top of descent — the **PRU** tops by default, see Top of climb and top of descent — with three exclusions: the climb floor, the descent floor, and the exclusion box that also relocates the tops.

16.2 The floors were being compared against the wrong datum

This is the second silent unit defect V4 closes, and it is the same class as the one in Airport surface events.

The climb floor is **3,000 ft AGL** (the end of the ICAO noise-abatement departure procedure) and the descent floor **1,800 ft AGL** (typical ILS interception). The shipped v0.1.0 code compared both against **uncorrected pressure altitude**. At Zurich, field elevation 1,416 ft, the descent floor therefore sat 1,416 ft too low; at Ibiza, field elevation 24 ft, it was effectively right. The result was a per-aerodrome bias in what counted as a level-off, with nothing in the output to show it. `level_floors_above_field` measures both against the field.

Parameters.

Parameter	Value	Source	Governs
`level method`	`pru`	ours	which arm runs
`level window seconds`	10 s	PRU	temporal size of the rolling window
`level vertical speed limit ftmin`	300 ft/min	PRU / ICAO	the Y/X ratio, and the ICAO arm's rate limit
`level window height ft`	50 ft	derived	the window's height at a 10 s window
`level band limit ft`	200 ft	ICAO	the ICAO arm's altitude band
`level min duration seconds`	20 s	PRU / ICAO	shortest reportable segment
`level min altitude climb ft`	3,000 ft AGL	PRU / ICAO	where climb analysis starts
`level min altitude descent ft`	1,800 ft AGL	PRU / ICAO	where descent analysis stops
`level floors above field`	on	ours	whether those two are AGL or above the 1013.25 hPa datum
`level analysis radius nm`	200 NM	PRU	radius around the aerodrome analysed
`level radius enforced`	on	ours	whether that radius binds at all; off reproduces v0.1.0, which declared it and read it nowhere
`level exclusion box pct`	90%	PRU / ICAO	lower edge of the cruise exclusion
`level exclusion box seconds`	300 s	PRU / ICAO	duration above which a high segment is cruise

None of the published values must be tuned. They are PRU's and ICAO's, and tuning them forfeits the only claim this family can make — conformance to a published methodology — in exchange for a number nothing can validate. `level_method`, `level_floors_above_field`, `level_radius_enforced` and `level_anchor` are ours and are choices about which published

Table 16.2

The level-arm comparison is not yet computed: the V4 benchmark chain has not been run against this checkout. See Provenance.

methodology to follow, not about its constants. `level_radius_enforced` is the odd one: it is not a choice anybody should make differently, but the radius binds a segment whose aerodrome the flight list could not name *out* of the analysis, so turning it on and off is the only way to say whether a change in the count came from the geometry or from the flight list's coverage — and it is the only way to reproduce v0.1.0, which declared the radius and never read it.

Table 16.1

Type	Events
`level-off-climb-start`	93,402
`level-off-climb-end`	93,402
`level-off-descent-start`	221,996
`level-off-descent-end`	221,996
`level-start`	406,994
`level-end`	326,599

Compared with reference data. None is possible, and this is the family where that matters most. No source available to OPDI holds level-segment truth. The claim is conformance, tested on synthetic trajectories: a climb at 280 ft/min is level under the PRU window and one at 320 ft/min is not, because at a 10 s sample the first rises 46.7 ft inside a 50 ft window and the second rises 53.3 ft. That is checkable; "accurate" is not.

Limitations. Unvalidated against data, by necessity. Two arms exist and neither can be shown better than the other, so the choice of default is an argument, not a measurement. The PRU arm's interpolation grid is built as a single array per partition and is capped, so a track whose segmentation failed and which spans a month loses the tail of its analysis — bounded and in the cheap direction, but real. And the arms mean slightly different things by `level_ft`, which a consumer comparing across `level_method` values would need to know.

17 Flight-level crossings

Types: `xing-fl50`, `xing-fl70`, `xing-fl100`, `xing-fl245`.

What it is. Every crossing of each monitored flight level, with the instant interpolated to the level itself.

How it is detected. The question is *when did the aircraft pass through this level*, and the reason it needs an algorithm rather than a comparison is noise.

A trajectory that sits near a threshold does not cross it once. Sampled at five seconds, an aircraft holding FL99.8 with fifty feet of barometric noise produces a crossing of FL100 every few samples, all of them spurious. The remedy is a **Schmitt trigger**: instead of one threshold there are two, a dead band either side of it. The signal is *above* only once it exceeds the upper edge and *below* only once it falls under the lower edge; between the two it is undefined and keeps whatever it last was. A crossing is published when that remembered side flips, so noise inside the band cannot produce one and a genuine crossing still does. The band here is `crossing_hysteresis_ft` = 300 ft.

The trigger decides *whether* a crossing happened. It does not decide *when*: the sample that confirms it is already past the threshold. The instant is obtained separately, by linear interpolation between the two samples that straddle the bare threshold — a different pair of rows from the one that confirmed it.

This is the engine four other families use. Airborne and Touchdown run it over height above field elevation, Landing over along-track distance from the threshold, Go-around over height at two levels, and Ring crossings over distance to an aerodrome. Each supplies its own quantity, its own thresholds and its own dead band; the mechanism is identical, which is why validating it in one place validates it in all of them.

Two further properties, both fixes to the published detector. Thresholds are evaluated as **parallel columns over one window** rather than by exploding the trajectory once per threshold, so N levels cost one shuffle rather than N — and that also removes a latent fault, because the published detector's first-match `when` chain reported only one level when a single 5 s interval spanned two. And an aircraft cruising *at* a level never leaves the dead band, so its side never resolves and **no event is emitted**, which is the property that makes reporting every crossing usable at all.

Parameters.

Parameter	Value	Source	Governs
`crossing levels fl`	50, 70, 100, 245	published vocabulary	which levels are monitored
`crossing hysteresis ft`	300 ft	ours	chatter suppression vs missed crossings
`crossing all occurrences`	on	ours	every crossing rather than first and last

Parameter	Value	Source	Governs
`crossing interpolate`	on	ours	interpolate to the level rather than report a sample

Each event carries `crossing_seq` (1-based, in time order), `direction`, and `bracket_s` — the spacing of the two interpolated-between samples, so a consumer can tell an instant measured across 5 seconds from one inferred across four minutes.

Table 17.1

Type	Events
`xing-fl50`	193,609
`xing-fl70`	200,191
`xing-fl100`	191,414
`xing-fl245`	160,424

Compared with reference data. None is possible — APDF holds no flight-level crossings. The detector's *mechanism* is nonetheless validated where the same code is applied to rings, which APDF does record: see Ring crossings.

Limitations. A flight that climbs and levels off **exactly at** a threshold never clears the far edge and registers no crossing there. That case is described by the level-segment and top-of-climb events instead, but it is a real semantic change from the published `first-xing-fl*` / `last-xing-fl*` family, which is retired — see Breaking changes for consumers.

18 Ring crossings

Types: `xing-40nm`, `xing-100nm`.

What it is. Crossings of the 40 NM and 100 NM cylinders around the flight's **own** origin and destination — ICAO's ASMA cylinder for KPI08 and the reference area for KPI05.

How it is detected. The Schmitt trigger and interpolation engine of Flight-level crossings, with distance to an aerodrome as the quantity and a 1 NM dead band.

The rings come from the flight list's ADEP and ADES rather than from the H3 aerodrome zone table: that is what the indicators mean, and what APDF records — one crossing per movement, not one per aerodrome overflown. It is also far cheaper, multiplying each sample by at most two rather than by every aerodrome within 110 NM.

Bearing is computed **from** the interpolated crossing position rather than interpolated alongside it. A bearing is circular: averaging 359 and 1 gives 180, wrong by half a turn on any crossing near due north.

Parameters.

Parameter	Value	Source	Governs
`ring radii nm`	40, 100 NM	ICAO	which cylinders
`ring hysteresis nm`	1 NM	ours	chatter suppression for a track flying tangentially
`crossing interpolate`	on	ours	shared with the level crossings

`info` carries `crossing_seq`, `direction` (inbound/outbound), `apt_icao`, `bearing` and `flight_level`, matching `C40_CROSS_{TIME,LAT,LON,FL}` and `C40_BEARING` column for column, so the comparison is direct rather than a translation.

Compared with reference data. This is the one family where the reference is second-resolution, so it is the one place a seconds-level claim is defensible. On V3's sample the detector's spread sat **inside the disagreement between two independent EUROCONTROL derivations of the same crossing** — median error under half a second, median position error about 0.2 NM. The V4 sample reproduces that: the table below is the ring comparison on these three days, and the sub-second bias is not a rounding artefact of the reference, because this is the one family whose reference resolves to the second.

Table 18.1

Ring	Coverage	Bias	MAD	Median position error	p90 position error
`xing-100nm`	94.96%	+0 s	2 s	0.17 NM	0.82 NM
`xing-40nm`	96.56%	+0 s	2 s	0.10 NM	0.60 NM

Limitations. A ring crossing is not detected for an aerodrome merely overflown — correct for these indicators, but not a general-purpose airspace crossing. And the family depends entirely on the flight list naming the aerodrome, so ADEP/ADES detection error propagates directly into it.

19 Airport surface events

Types: entry- / exit- for taxiway, apron, parking_position, threshold, hangar, deicing_pad.

What it is. Traversals of the aerodrome's surface features, from OpenStreetMap aeroway geometry.

entry-runway and exit-runway are no longer part of this family. They were the one feature where a traversal's *purpose* mattered, and they are replaced by the classified runway family — Line-up through Runway crossings. See Breaking changes for consumers.

How it is detected. Low-altitude samples are matched against OSM aeroway geometry pre-rasterised to H3 resolution 12 in hexaero_airport_layouts. A sample joins a feature when its h3_res_12 cell matches *and* the feature's aerodrome is one the flight list names for that flight. Consecutive matches on the same feature form a traversal; a gap over 300 s starts a new one. Each traversal yields an entry and an exit event.

19.1 The gate was measured against the wrong datum

The published gate is **FL20 against the 1013.25 hPa datum**. baro_altitude_c is uncorrected pressure altitude, so on a low-pressure day, or at any aerodrome high enough, the gate moves relative to the ground it is supposed to sit just above. It is the same defect class as the phase family's ground ceiling, and the same one the level floors carry in Level segments: a threshold whose *intent* is "near the ground" compared against a datum that is not the ground.

airport_gate_above_field measures the gate as **2,000 ft above field elevation**, which is FL20's intent expressed against the field. The gate runs before the column projection narrows the frame, because the height calculation needs the raw altitude and the two elevation columns, none of which survive the select.

Parameters.

Parameter	Value	Source	Governs
airport gate above field	on	ours	whether the gate is AGL or against the 1013.25 datum
airport max height agl ft	2,000 ft	ours	the gate's value when it is AGL
airport max fl	FL20	published	the gate's value when it is not
airport trace gap seconds	300 s	published	when a traversal is considered ended
airport events ordered	on	ours	attributes taken at the entry/exit *times*
H3 resolution	12	H3Config	matching granularity

`airport_events_ordered` is a correctness fix, not a tuning knob: the published detector used `first/last` inside a `groupBy`, which take **partition order, not time order**, so a reported entry position was not guaranteed to be the position at the reported entry time.

Table 19.1

Feature	Entries	Exits
`taxiway`	25,220	24,553
`apron`	3,849	3,844
`parking_position`	4,051	4,015
`threshold`	87	87
`hangar`	10	10
`deicing_pad`	—	—

Compared with reference data. None is possible for any of them. APDF records a runway designator (`AP_C_RWY`) and a stand (`AP_C_STND`), but **no entry or exit times for any surface feature**. A runway *identity* can be checked indirectly through Touchdown; no aeroway event's *timing* can be checked at all.

Limitations. Cannot fire for an aerodrome the flight list did not name. Depends on OSM coverage, which is built only for large and medium aerodromes in the ingestion area. A traversal completed inside a single 5 s sample is dropped, because its duration is zero. Threshold polygons are the smallest features in the set and therefore the most sensitive to cleaning removing surface samples — V3 measured the largest relative fall of any event type there, and it was trajectory cleaning, not a detector change. `deicing_pad` is in the vocabulary but was not observed on V3's sample, because de-icing pads are tagged in OSM for few of the aerodromes covered and the sample is June.

20 First seen and last seen

Types: `first_seen`, `last_seen`.

What it is. The first and last sample of each track.

How it is detected. Row number ascending and descending over the track, ordered by time. Emitted for every track without condition, so the count is also the track count.

Parameters. None.

Compared with reference data. None is possible, and not for want of a source. These are properties of *reception coverage*, not of the flight. There is nothing they could be right or wrong against — an aircraft is first seen when a receiver first hears it, which no reference dataset records because it is not a property of the operation.

Limitations. May carry a null position where the first sample has none. Their value to a consumer is as a **coverage diagnostic**: the gap between `first_seen` and Off-block and on-block, or between `last_seen` and Runway vacated, is the part of the movement ADS-B did not see, and in this study set it is smaller than it is anywhere else in the network.

Table 20.1

`first_seen` **181,721**, `last_seen` **181,721** — which is also the number of tracks in the sample.

21 Measurements

Two values attached to every event rather than being events themselves.

`Distance flown (NM)` is the cumulative great-circle distance from the start of the track, summed sample to sample with an Earth radius of 6,371 km and converted at 1.852 km per nautical mile. `Time Passed (s)` is elapsed time from the track's first sample.

Both are **interpolated to the crossing instant** for every family that produces an interpolated instant — the crossings, the rings, `airborne`, `touchdown` and `landing`. This is not a refinement: a measurement attached to an interpolated event but read from a neighbouring row would disagree with its own event, and the disagreement would be invisible in the published table.

Compared with reference data. APDF carries `C40_TRANSIT_DIST_KM` — the achieved distance inside the ring — which makes a distance comparison possible for the ring families. **It has not been run**, in V3 or here; it remains the most tractable unmeasured quantity in this programme.

Limitations. Cumulative from the *track's* start, not from the flight's, so a flight whose track begins mid-climb has a distance measure that is not the distance flown. Track identity changed shape with the A8 segmentation default, so these measures are not comparable across that boundary either.

22 Cross-cutting findings

Six results are about the programme rather than about any one milestone.

22.1 A published take-off can be shadowed by a level-start

The legacy phase family emits its events from a single `when` chain, and the `take-off` arm sits **after** the level-segment arms in it. A sample the fuzzy classifier labels `LVL` therefore publishes a `level-start` and the take-off at that same sample is never reached.

This is pre-existing behaviour, unchanged by V4, and it is narrow — it needs the rotation sample to be classified level rather than climb, which happens when the vertical rate is still small and the smoothing window has not yet tipped. It surfaced while building a test fixture, not from the scored metrics, which is the point worth recording: **an ordering sensitivity inside a `when` chain produces a missing event, and a missing event looks exactly like a detector that did not fire.** It is one more reason the A-CDM family derives its milestones from an explicitly classified object rather than from the order of arms in a chain.

22.2 A circular quantity is being averaged linearly, and it fails silently

The runway direction is chosen using the traversal's **median ground track**, computed as `percentile_approx(heading, 0.5)`. Heading is circular and the median is not: a runway near bearing 360 whose samples straddle 359° and 001° yields a median near 180° — the exact reciprocal.

The alignment fold to [0°, 90°] hides this from the *classification*, because a reciprocal still counts as aligned, so the traversal is still called an arrival and `touchdown` still fires. But Landing is measured as a sign change of along-track distance **from the chosen threshold**, and measured from the wrong end of the strip that sign change never occurs inside the traversal. The failure mode is therefore **a silently missing `landing` on runways numbered near 36/18**, with a `touchdown` present on the same traversal to make the gap look like a detector quirk rather than a defect.

It is inherited from the shipped `detect_runway_movements` rather than introduced here, and fixing it means a circular median — a deliberate deviation from the reference implementation, which is why it was not done inside this change. The check that would settle it is a cut of the runway-identity table by designator, and it is the first thing to run when the chain completes.

22.3 Two vocabularies wrote the runway designator under two keys

`ATOT/ALDT` recorded the runway in `info` under `runway`; the A-CDM family records it under `rwy_ident`. Both readers in the benchmark parsed only `runway`, so **every v0.2.0 designator would have read as NULL — indistinguishable from "the detector named no runway"**, and the runway-identity result would have collapsed to zero with a plausible-looking explanation available.

It was found by wiring the comparison, not by a test, and both readers now coalesce the two keys. It is recorded here because it is the characteristic failure of a version bump: not a wrong number, but a right number read from a key that moved.

22.4 Abstention is an output, not a gap

Three places in the V4 detectors decline to answer rather than guessing: the 30°–45° traversal band, the runway pick when no runway is within tolerance, and a NULL distance failing the 200 NM radius test rather than passing it.

Each costs coverage and each is deliberate. **Naming the wrong milestone corrupts a movement count; naming none is an explicit null a consumer can see.** Every coverage figure in this document is therefore lower than it could be made by a detector willing to guess, and that trade is the reason the figures can be read at all.

22.5 The suppression floor, and why the twenty are not comparable

The twenty study aerodromes differ in size by a factor of about twenty over the three-day sample. A percentile computed on a handful of detections is noise wearing a number's clothes, so Annex A suppresses any cell with fewer than 20 detections — **marked, not dropped**, because an aerodrome removed from a table is indistinguishable from one that was never in the study.

22.6 Two negative results from V3 still stand

Neither was re-tested here and neither has been reverted.

Requiring complete fuzzy evidence made detection worse, on both V3 periods. Spark's `least/greatest` skip nulls, so a rule missing an input became a minimum over fewer terms and could out-compete a complete rule. Requiring every input is obviously more correct, and it removed most phase-based landings, because ADS-B carries a null velocity often enough that insisting on complete evidence discards most ground samples. It matters less than it looks now that the runway-anchored family replaces that path entirely.

The 60-second smoothing window was contradicted by the second period. It is what OpenAP does and what the original port dropped, so it has the strongest prior justification of any change in the programme; it was near-neutral on one sample and cost 86% of landing coverage on the other. One confound is stated with it: the second period's tracks predate the altitude repair, so input preparation differed as well as the month. The actionable half is that a fix behaving this differently on noisier input should not be adopted on the strength of its rationale alone.

23 Breaking changes for consumers

The new version string is `events_v0.2.0`. `EventConfig.legacy()` still reproduces `events_v0.0.2` exactly, and the benchmark's baseline rung reproduces `events_v0.1.0`, so **any past month remains reproducible** under the configuration that produced it.

23.1 `landing` has changed meaning, and kept its name

This is the one change that can corrupt an existing consumer silently, so it leads.

	Under `events_v0.0.2` / `v0.1.0`	Under `events_v0.2.0`
Question	did the fuzzy phase go descent → ground	when did the aircraft cross the landing threshold
ICAO	—	**T16**
Instant	a sampled state vector	interpolated to the threshold plane
Scored against	APDF ALDT	**nothing** — ALDT is Touchdown's

A consumer reading `landing` as touchdown will now be reading the threshold crossing, which is earlier by the flare and the distance from threshold to touchdown point. **Two things distinguish them in the published data**: the `version` column, and `info.milestone`, which is `T16` under the new meaning and absent under the old. Consumers wanting touchdown must move to `touchdown`.

23.2 Types no longer emitted under `events_v0.2.0`

Each is superseded, not deleted: it remains reachable under the configuration that published it, and each line points at the chapter that documents its successor.

Retired type	Replaced by	Why
`take-off`	Airborne (T08)	fuzzy phase transition → interpolated 15 ft AGL crossing on a classified departure. V3 measured `take-off` at 0.39% of departures; `airborne` reaches 76.6% here. The two were not run on the same sample, so the gap is indicative rather than a like-for-like result — but it is three orders of magnitude, not a margin
`AOBT`	Off-block and on-block (T04)	renamed to `off-block`; same detector
`AIBT`	Off-block and on-block (T21)	renamed to `on-block`; same detector
`entry-runway`	Line-up, Runway crossings	one type conflated departures, arrivals and taxi crossings
`exit-runway`	Runway vacated, Runway crossings	same

Already retired at `events_v0.1.0` and still not emitted: `first-xing-fl{50,70,100,245}` and `last-xing-fl{50,70,100,245}`, replaced by the crossing sequence number in Flight-level crossings — the first crossing is `crossing_seq == 1`, the last is the maximum.

23.2.1 `ATOT/ALDT` are retained, not retired

The A-CDM `airborne/touchdown` were designed to *replace* `ATOT/ALDT`. On coverage they do not: measured on this sample `ATOT` answers 89.8% of reachable departures against `airborne`'s 76.6%, and `ALDT` 95.9% against `touchdown`'s 80.4%. On timing the new pair is the better of the two — at the three aerodromes where APDF resolves to the second the takeoff instant lands within a few seconds of truth, against `ATOT`'s +25.0 s on this sample (+19 s as V3 measured it on 2025). That per-aerodrome figure pools both detectors, since the per-aerodrome table carries no detector column, so attributing the few-second result to `airborne` is an inference from `ATOT`'s much larger bias rather than a direct per-detector measurement. The two methods read the aircraft at different heights (see Timing a runway movement: two methods, four milestones): `ATOT/ALDT` in the climb and descent up to 1 500 ft, `airborne/touchdown` at the 15 ft crossing on and beside the runway. The second needs surface and low-altitude reception the first does not. Rather than surrender the older detectors' coverage for the newer ones' accuracy, `events_v0.2.0` **publishes both**, distinguished by type string, so a consumer chooses the property it needs. A row-count alarm keyed on `ATOT/ALDT` disappearing would therefore be a false alarm — they are still emitted.

23.3 Other changes worth a row-count alarm

`info` keys moved. The runway designator is `rwy_ident` in the A-CDM family, where it was `runway` for `ATOT/ALDT`. A consumer parsing `info` should coalesce the two rather than switching on version.

The event table changes size in both directions. Nine new runway types and the PRU tops add events; the retirements and the level-arm change remove them. Volume is not a proxy for information here, and any consumer with a row-count alarm on the event table should expect it to fire.

Every top of climb and descent now names its algorithm in `info.method` (`phase` or `pru`), and four type names exist where two did. A consumer selecting `type = 'top-of-climb'` gets the fuzzy pair only, unchanged; one wanting PRU's must ask for `top-of-climb-cco`.

Event identifiers are derived from the event's own identity — track, type, time and version — rather than from a partition counter, so a re-run produces the same identifiers. This makes a re-run idempotent in *content*; it does not make the write idempotent, and re-processing safety still rests on the per-month progress logs.

Track identity changed shape independently of this version. The A8 segmentation default removed the `_{year}_{month}` suffix from `track_id`, so any consumer parsing that suffix breaks, and past months do not reproduce. That is a track-construction change, documented in its own paper, and it is mentioned here only because it lands in the same release window.

24 Limitations

The ones that bound how this document should be read, collected.

The study set is the best-covered twenty aerodromes, chosen on a ranking computed over the same three days that are scored. Every figure is an upper bound, and a detector limited by reception looks better here than anywhere else in the network.

Most reference times are minute-rounded. A bias under a minute at such an aerodrome cannot be distinguished from APDF's own quantisation. Annex A carries the resolution per aerodrome for exactly this reason.

The reference join has a ceiling — `ID` was null on 2.83% of APDF rows on the June 2025 extract, the share for this period is measured in Ground truth, and its ceiling, and the reach rate to `icao24` caps every coverage figure above that.

Ten milestone families have no external truth at all: (1) the tops of climb and descent, (2) both level-segment arms, (3) the flight-level crossings, (4) every airport surface event, (5) `line-up`, (6) `take-off-roll`, (7) `landing` as an instant — ALDT belongs to `touchdown` — (8) `runway-vacated`, (9) the runway crossings and (10) `go-around`. For the level family and the PRU tops the substitute is conformance to a published methodology, tested on synthetic geometry. For the others there is none, and their counts are reported while their timing is unvalidated and stated as such.

Five tuning parameters are module constants, not config fields — `ROLL_MAX_GAP_SECONDS`, `AIRBORNE_HYSTERESIS_FT`, `THRESHOLD_PLANE_HYSTERESIS_NM`, `GOAROUND_HYSTERESIS_FT` and `ARRIVAL_LEAD_SECONDS`. They are documented where they sit, but they are not sweepable without editing code, which contradicts the convention every other threshold in this document follows.

Known defects carried, not fixed: the circular median described in Cross-cutting findings; the `when`-chain ordering sensitivity in the legacy phase family; the PRU arm's interpolation-grid cap, which truncates the tail of a pathologically long track; and the non-deterministic go-around low point on a flat minimum.

No parameter sweep has been run. Every threshold is named in `EventConfig`, which is what makes one possible, and V3's chapter on designing one — an objective that trades coverage against error, tolerances taken from the inter-source floor rather than chosen, fit on one period and validated on the other — still describes the work that has not been done.

25 Provenance

Rendered with OPDI_RENDER=allow-stale: the regeneration chain was not run, so every figure below is whatever was last staged under data/.

Every figure on this page is produced by `benchmarks/regenerate_events_v4.py`, which reruns a job only when the source files that job depends on have changed since the output was written. **An output with no manifest entry below is reported in the text as not yet computed rather than shown as fact.**

Table 25.1

Output	Script	Git SHA	Produced
`bridge_2026.json`	`benchmarks/events_gt.py`	`b55ae43`	2026-09-03
`inventory_2026.csv`	`benchmarks/event_bench.py`	`39093a1`	2026-09-08
`ladder_2026.csv`	`benchmarks/event_bench.py`	`39093a1`	2026-09-08
`per_airport_2026.csv`	`benchmarks/events_compare.py`	`39093a1`	2026-09-08
`per_airport_by_detector_2026.csv`	`benchmarks/events_compare.py`	`39093a1`	2026-09-08
`pooling_rules_2026.csv`	`benchmarks/events_compare.py`	`39093a1`	2026-09-08
`resolution_2026.csv`	`benchmarks/events_compare.py`	`39093a1`	2026-09-08
`rings_2026.csv`	`benchmarks/events_compare.py`	`39093a1`	2026-09-08
`table:s3a://eurocontr`[illegible]	`/home/jupyter/work/`[illegible]`opdi/.claude/worktrees/flight-events-`	`b55ae43`	2026-09-03

Table 25.2

Expected output	Staged
`ladder_2026.csv`	yes
`per_airport_2026.csv`	yes
`runway_2026.csv`	yes
`resolution_2026.csv`	yes
`inventory_2026.csv`	yes
`bridge_2026.json`	yes

Four caveats belong with the provenance rather than with any figure.

The staleness check is not credential-free. `provenance.inputs_changed` reports "not stale" when it cannot reach S3 to look, so a check run without credentials is not evidence of currency. The stage that builds the 2026 flight list reports `cannot check (no credentials)` — i.e. stale — rather than passing, which is the correct direction, but the job-level check does not have that property.

The reference-join ceiling and the scored population are different scopes. The ceiling is computed over the movements the bridge could reach; the scores are computed over the study set on three days. Reading the first as the denominator of the second overstates coverage.

The reference extract is a month; the sample is three days. The scorer restricts truth to the benchmark days before scoring. This is stated twice in this document because getting it wrong produces a plausible table.

V3's staged outputs now read stale against this code. V4's changes touch `event_bench.py`, which is in V3's dependency set, so V3's fingerprints moved even though its committed CSVs did not. V3's numbers are traceable to the version that published them and were not re-run.

Annex A — Accuracy by aerodrome

One table per milestone, one row per aerodrome, in coverage-ranking order — EBBR first, UGKO twentieth. The columns are the same throughout:

- **n truth** — reference movements for that aerodrome and milestone, over the three sample days, after the bridge.
- **Coverage** — the share of those with a detection aligned to them.
- **Bias** — median signed error, detection minus reference. Positive is late.
- **MAD** — median absolute error.
- **p90** — 90th percentile of absolute error.
- **within 60 s** — share of *truth* rows detected within a minute.
- **Truth resolution** — `second` where most of that aerodrome's reference times carry a non-zero seconds field, `minute` otherwise.

Read the last column first. At a `minute` aerodrome a bias under 30 s is inside the reference's own quantisation and means nothing; at a `second` aerodrome it is a measurement. Comparing an aerodrome of one kind against one of the other without that column is the specific mistake this annex exists to prevent.

Dashes are **suppressed cells, not missing detections**: fewer than 20 detections, below the floor at which a percentile carries information. The detections behind each dash are listed under the table, so a suppressed row can still be told apart from an absent one.

Every figure here is measured against APDF, not against another OPDI milestone: `n truth` counts APDF movements at that aerodrome and the bias is the detected instant minus the APDF one. The section headings said "against `airborne`" in an earlier draft, which read as though one OPDI milestone were being scored against another; nothing of the sort is computed anywhere in this paper.

Each row pools every detector that answers the milestone. Under `events_v0.2.0` the ATOT milestone is answered by both `ATOT` and `airborne`, and the ALDT milestone by both `ALDT` and `touchdown`, so a row is "what OPDI published for this movement, whichever detector produced it" — the question a consumer of the table actually has. It is *not* a per-detector figure, because the scorer groups on (aerodrome, milestone) and carries no detector column. For the per-detector split, network-wide, see What the evidence says; the per-detector split *per aerodrome* is not measured by this campaign.

A.1 ATOT — against APDF

Aerodrome	n truth	Coverage	Bias	MAD	p90	within 60 s	Truth resolution
EBBR	843	97.86%	+31 s	32 s	55 s	94.90%	minute
LSZH	1,186	98.31%	+4 s	4 s	7 s	97.89%	second

Aerodrome	n truth	Coverage	Bias	MAD	p90	within 60 s	Truth resolution
EICK	121	91.74%	+35 s	35 s	65 s	80.17%	minute
EFHK	638	93.89%	+50 s	50 s	75 s	61.76%	minute
LEIB	517	95.36%	-2 s	4 s	12 s	95.16%	second
EDDS	371	97.04%	+2 s	3 s	9 s	96.50%	second
LFLL	368	98.37%	+35 s	35 s	60 s	92.39%	minute
LHBP	607	95.06%	+1 s	15 s	29 s	94.40%	minute
LFPO	972	96.60%	+20 s	20 s	45 s	96.60%	minute
LPFR	395	98.23%	+35 s	35 s	60 s	89.62%	minute
ESSA	886	91.08%	+49 s	50 s	75 s	65.24%	minute
ENVA	199	91.96%	+25 s	30 s	65 s	79.40%	minute
LOWW	1,070	96.36%	+25 s	25 s	50 s	95.61%	minute
LKPR	664	97.29%	+9 s	14 s	30 s	96.69%	minute
EDDP	243	95.47%	+20 s	20 s	50 s	94.65%	minute
LPPT	993	98.09%	+40 s	40 s	65 s	81.17%	minute
EGGD	370	96.49%	+30 s	30 s	55 s	94.59%	minute
EGNT	215	93.02%	+30 s	30 s	55 s	92.09%	minute
EGCC	973	97.12%	+30 s	35 s	59 s	94.96%	minute
UGKO	44	90.91%	+0 s	14 s	29 s	90.91%	minute

A.2 ALDT — against APDF

Aerodrome	n truth	Coverage	Bias	MAD	p90	within 60 s	Truth resolution
EBBR	823	97.21%	+10 s	11 s	37 s	96.96%	minute
LSZH	1,198	96.91%	-5 s	5 s	9 s	96.49%	second
EICK	120	96.67%	+20 s	30 s	50 s	95.83%	minute
EFHK	631	93.50%	+4 s	13 s	30 s	92.87%	minute
LEIB	507	97.63%	-7 s	7 s	10 s	97.04%	second
EDDS	375	97.60%	-13 s	13 s	52 s	90.40%	second
LFLL	374	98.40%	+6 s	10 s	29 s	98.13%	minute
LHBP	612	95.59%	-10 s	13 s	35 s	94.28%	minute
LFPO	977	96.52%	+21 s	22 s	50 s	96.11%	minute
LPFR	393	97.96%	-5 s	10 s	30 s	97.71%	minute
ESSA	884	93.33%	+2 s	15 s	30 s	92.53%	minute
ENVA	199	96.98%	+5 s	20 s	50 s	89.45%	minute
LOWW	1,068	97.10%	+5 s	10 s	28 s	96.72%	minute
LKPR	668	95.66%	+19 s	19 s	48 s	94.76%	minute
EDDP	223	97.31%	+15 s	20 s	45 s	96.41%	minute
LPPT	994	97.28%	-5 s	9 s	30 s	97.18%	minute
EGGD	378	93.65%	+15 s	16 s	45 s	92.59%	minute
EGNT	212	91.04%	+21 s	25 s	50 s	90.09%	minute
EGCC	981	93.58%	+20 s	20 s	45 s	92.86%	minute
UGKO	40	85.00%	-5 s	15 s	45 s	85.00%	minute

A.3 AOBT — against APDF

Aerodrome	n truth	Coverage	Bias	MAD	p90	within 60 s	Truth resolution
EBBR	845	—	—	—	—	—	minute
LSZH	1,185	63.97%	+722 s	725 s	1048 s	0.00%	second
EICK	121	—	—	—	—	—	minute
EFHK	640	—	—	—	—	—	second
LEIB	513	—	—	—	—	—	minute
EDDS	367	42.51%	+705 s	705 s	1005 s	0.00%	minute
LFLL	368	—	—	—	—	—	minute
LHBP	609	—	—	—	—	—	second
LFPO	972	—	—	—	—	—	minute
LPFR	397	—	—	—	—	—	minute
ESSA	885	—	—	—	—	—	minute
ENVA	199	—	—	—	—	—	minute
LOWW	1,071	2.15%	+600 s	675 s	3615 s	0.00%	minute
LKPR	661	4.84%	+765 s	770 s	3040 s	0.00%	minute
EDDP	243	—	—	—	—	—	minute
LPPT	994	—	—	—	—	—	minute
EGGD	370	—	—	—	—	—	second
EGNT	215	—	—	—	—	—	minute
EGCC	975	—	—	—	—	—	minute
UGKO	44	—	—	—	—	—	minute

Dashes are suppressed cells, not missing detections: fewer than 20 detections, below the floor at which a percentile means anything. Detections behind each dash — EBBR (10 detected); EICK (1 detected); EFHK (10 detected); LEIB (8 detected); LFLL (3 detected); LHBP (17 detected); LFPO (1 detected); LPFR (4 detected); ESSA (7 detected); ENVA (0 detected); EDDP (9 detected); LPPT (12 detected); EGGD (2 detected); EGNT (0 detected); EGCC (7 detected); UGKO (2 detected).

A.4 AIBT — against APDF

Aerodrome	n truth	Coverage	Bias	MAD	p90	within 60 s	Truth resolution
EBBR	825	—	—	—	—	—	minute
LSZH	1,197	55.97%	-300 s	302 s	482 s	0.08%	second
EICK	120	—	—	—	—	—	minute
EFHK	633	—	—	—	—	—	second
LEIB	506	—	—	—	—	—	minute
EDDS	375	23.47%	-393 s	457 s	12253 s	0.00%	second
LFLL	374	—	—	—	—	—	minute
LHBP	611	—	—	—	—	—	second
LFPO	977	—	—	—	—	—	minute

Aerodrome	n truth	Coverage	Bias	MAD	p90	within 60 s	Truth resolution
LPFR	395	—	—	—	—	—	minute
ESSA	882	—	—	—	—	—	minute
ENVA	199	—	—	—	—	—	minute
LOWW	1,069	2.71%	-290 s	290 s	2775 s	0.00%	minute
LKPR	667	3.45%	-210 s	210 s	510 s	0.00%	minute
EDDP	223	—	—	—	—	—	minute
LPPT	994	—	—	—	—	—	minute
EGGD	378	—	—	—	—	—	minute
EGNT	212	—	—	—	—	—	minute
EGCC	980	—	—	—	—	—	minute
UGKO	40	—	—	—	—	—	minute

Dashes are suppressed cells, not missing detections: fewer than 20 detections, below the floor at which a percentile means anything. Detections behind each dash — EBBR (14 detected); EICK (1 detected); EFHK (13 detected); LEIB (5 detected); LFLL (3 detected); LHBP (17 detected); LFPO (5 detected); LPFR (7 detected); ESSA (12 detected); ENVA (0 detected); EDDP (6 detected); LPPT (10 detected); EGGD (2 detected); EGNT (0 detected); EGCC (7 detected); UGKO (3 detected).

A.5 Per-detector accuracy by aerodrome

A.1–A.4 pool the detectors that answer a milestone: the detection nearest the reference wins and the other is discarded, so a row is *what OPDI published for that movement*. These tables keep the detectors apart. Both sides are scored against the same APDF truth and, because a detector that produced nothing for a movement is still charged that movement, the two rows of a pair are read down the column against each other.

The two questions have different answers, and this is where that shows.

Aerodrome	n truth	ATOT coverage	ATOT bias	airborne coverage	airborne bias	Truth resolution
EBBR	843	95.26%	+30 s	91.81%	+36 s	minute
LSZH	1,185	97.81%	+4 s	97.81%	+7 s	second
EICK	121	91.74%	+35 s	7.44%	+38 s	minute
EFHK	638	85.74%	+50 s	46.64%	+51 s	minute
LEIB	516	71.51%	-6 s	94.20%	-1 s	second
EDDS	372	97.04%	+1 s	95.42%	+6 s	second
LFLL	368	98.10%	+35 s	98.10%	+41 s	minute
LHBP	607	89.79%	+0 s	85.71%	+5 s	minute
LFPO	972	96.40%	+20 s	91.36%	+23 s	minute
LPFR	395	96.71%	+35 s	97.47%	+38 s	minute
ESSA	886	90.63%	+50 s	45.02%	+46 s	minute
ENVA	199	88.94%	+25 s	21.61%	+38 s	minute
LOWW	1,070	50.65%	+20 s	92.05%	+26 s	minute
LKPR	664	96.54%	+10 s	94.13%	+11 s	minute

Aerodrome	n truth	ATOT coverage	ATOT bias	airborne coverage	airborne bias	Truth resolution
EDDP	243	95.06%	+20 s	64.61%	+23 s	minute
LPPT	993	97.89%	+40 s	97.38%	+45 s	minute
EGGD	370	96.22%	+30 s	35.95%	+33 s	minute
EGNT	215	93.02%	+30 s	16.28%	+22 s	minute
EGCC	974	96.92%	+30 s	33.09%	+33 s	minute
UGKO	44	90.91%	+0 s	90.91%	+3 s	minute

Aerodrome	n truth	ALDT coverage	ALDT bias	touchdown coverage	touchdown bias	Truth resolution
EBBR	823	97.08%	+20 s	90.28%	+8 s	minute
LSZH	1,198	96.91%	-5 s	90.54%	-17 s	second
EICK	120	96.67%	+20 s	33.33%	+21 s	minute
EFHK	631	93.50%	+10 s	79.27%	+1 s	minute
LEIB	508	97.64%	-7 s	97.44%	-19 s	second
EDDS	378	97.35%	-13 s	80.59%	-26 s	second
LFLL	374	98.40%	+15 s	88.77%	+4 s	minute
LHBP	611	95.58%	-10 s	92.66%	-26 s	minute
LFPO	977	96.42%	+30 s	96.42%	+22 s	minute
LPFR	392	97.96%	-5 s	97.22%	-11 s	minute
ESSA	884	93.33%	+5 s	57.53%	-2 s	minute
ENVA	199	95.98%	+10 s	36.68%	+7 s	minute
LOWW	1,068	97.10%	+15 s	93.65%	+1 s	minute
LKPR	668	95.66%	+35 s	92.51%	+19 s	minute
EDDP	223	97.31%	+25 s	91.48%	+15 s	minute
LPPT	994	97.28%	+0 s	96.88%	-19 s	minute
EGGD	378	93.65%	+20 s	59.52%	+14 s	minute
EGNT	212	91.04%	+25 s	27.36%	+23 s	minute
EGCC	979	93.56%	+20 s	30.31%	+17 s	minute
UGKO	40	85.00%	-5 s	82.50%	-16 s	minute

What these two tables say. On departures `airborne` beats `ATOT` on coverage at **four of the twenty** aerodromes, and where it does the gap is not marginal: LOWW **92.1% against 50.7%**, LEIB **94.2% against 71.5%**. At the other sixteen `ATOT` is ahead. On arrivals `touchdown` beats `ALDT` at **none** of the twenty.

That is the whole case for publishing both, stated precisely. It is not that the A-CDM family is better; on this sample it is not. It is that `ATOT` collapses at a small number of aerodromes and `airborne` does not collapse at the same ones, so the pooled figure is worth **+6.5 points** on departures. On arrivals `ALDT` is ahead everywhere and pooling gains **+0.04 points** – `touchdown` earns its place in the table on its A-CDM semantics and its 4D fix, not on adding answers a consumer could not otherwise get.